

The determinant of the A_t is

$$D = \begin{vmatrix} 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & -6 & 12 & -20 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6 & 12 & 20 \end{vmatrix} = 0$$

To prove that this is zero, we expand by the elements of the second row and then the elements of the fifth row

$$D = (-1)(-2) \begin{vmatrix} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 0 & -6 & 12 & -20 \\ 0 & 6 & 12 & 20 \end{vmatrix} = 0$$

Now add the second row to the first and the fourth to the third to get

$$D = 2 \begin{vmatrix} 0 & 0 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 24 & 0 \\ 0 & 6 & 12 & 20 \end{vmatrix} = 0$$

The first and third rows are proportional to each other, and so the determinant is 0.

Thus there is no interpolating polynomial. In a sense the data put four conditions on the even part and two on the odd part of the polynomial.

However, there are infinitely many formulas of the form

$$\begin{aligned} \int_{-1}^1 f(x) dx = & a_{-1}f(-1) + a_0f(0) + a_1f(1) \\ & + b_{-1}f''(-1) + b_0f''(0) + b_1f''(1) \end{aligned}$$

which use the same data, and the formulas are exact for quintics. To show this, we write the defining equations

$$\begin{aligned} 1: \quad 2 &= a_{-1} + a_0 + a_1 \\ x: \quad 0 &= -a_1 + a_1 \\ x^2: \quad \frac{2}{3} &= a_{-1} + a_1 + 2(b_{-1} + b_0 + b_1) \\ x^3: \quad 0 &= -a_{-1} + a_1 + 6(-b_{-1} + b_1) \\ x^4: \quad \frac{2}{5} &= a_{-1} + a_1 + 12(b_{-1} + b_1) \\ x^5: \quad 0 &= -a_{-1} + a_1 + 20(-b_{-1} + b_1) \end{aligned}$$

Using a_1 as a parameter (since we know from immediately above that the determinant has rank 5), we get

$$\begin{aligned} \int_{-1}^1 f(x) dx = & 2f(0) + \frac{1}{60} [f''(-1) + 18f''(0) + f''(1)] \\ & + a_1 \{ [f(-1) - 2f(0) + f(1)] \\ & - \frac{1}{12} [f''(-1) + 10f''(0) + f''(1)] \} \end{aligned}$$

If we impose the condition that the formula be exact for x^6 as well, then we get

$$\begin{aligned} \frac{2}{7} &= \frac{1}{60} (30 + 30) + a_1 \left[(1 + 1) - \frac{1}{12} (30 + 30) \right] \\ \frac{2}{7} &= 1 + a_1(2 - 5) \end{aligned}$$

or

$$a_1 = \frac{5}{21}$$

Thus we have

$$\begin{aligned} \int_{-1}^1 f(x) dx = & \frac{1}{21} [5f(-1) + 32f(0) + 5f(1)] \\ & - \frac{1}{315} [f''(-1) - 32f''(0) + f''(1)] \end{aligned}$$

which is exact for sixth degree polynomials.

The complete explanation of this apparent paradox is nontrivial,¹ but it should be no surprise that occasionally the two-step process [finding the interpolating polynomial and then applying the operator (integration from -1 to $+1$ in this case)] fails even though the direct one-step process succeeds. What is surprising is that it happens in this "reasonable" case.

PROBLEMS

17.7.1 Discuss the importance of

$$Q(x) = 3x^5 - 10x^3 + 7x$$

[which has $Q(-1) = Q(0) = Q(1)$ and $Q'(-1) = Q'(0) = Q'(1)$] to the example of noninterpolatory polynomials.

17.7.2 Give another example of a noninterpolatory polynomial.

17.7.3 If the integration was from 0 to 1 instead of -1 to 1, show that there is no such integration formula.

¹ M. P. Epstein and R. W. Hamming, "Non-Interpolatory Quadrature Formulas," SIAM.

17.8 AN EXPERIMENT IN COMPARING THE VALUE OF DERIVATIVES

The existence of the inverse determinant in the Birkhoff interpolation case shows that the use of *consecutive* derivatives at a point (omitting none avoids the example in Sec. 17.7) gives about as much information as only function values at points; the linear independence of the powers of x is preserved. Or, as in information theory, a polynomial of degree N can be reconstructed from the $N + 1$ Birkhoff samples.

Can we be more precise in this comparison of more sample points versus more derivatives at the old sample points? One approach is to examine the size of the error terms for various families of formulas. Assuming that the $G(s)$ function is of constant sign, then the E_m would provide one measure of the relative merit—though let us remember we are assuming that over the unspecified ensemble of functions we are considering, the value of the θ does not favor one formula over another.

We shall use the integral

$$\int_{-1}^1 f(x) dx$$

as the basis for an experiment.

The E_m can be found by integrating *any* polynomial of degree m with leading coefficient 1. By picking a polynomial with zeros at the sample points (meaning a multiplicity at x_i great enough to make the highest derivative used at that point zero at that point) we do not have to consider the right-hand side of the formula since all the values will then be zero. When the degree of the polynomial is odd, then the formula is clearly accurate for a power one higher, and we must add another factor to the polynomial we are using. It is simplest to pick another factor x to do this. An example of this effect is Simpson's formula which is accurate for one higher power of x than it was designed for.

The first family we shall examine is the one-point family

$$\int_{-1}^1 f(x) dx = w_0 f(0) + w_1 f'(0) + w_2 f''(0) + \cdots + w_n f^{(n)}(0)$$

For N an even number $E_{N+1} = 0$ from symmetry; hence we need only examine the odd cases for N , and m is even. We have

$$E_{2p} = \int_{-1}^1 x^{2p} dx = \frac{2}{2p+1}$$

In particular

$$E_2 = 2/3 \quad E_4 = 2/5 \quad E_6 = 2/7 \quad E_8 = 2/9 \quad E_{10} = 2/11$$

A natural second case is the family of two-point formulas

$$\int_{-1}^1 f(x) dx = \sum_{k=0}^N [w_{-1}^{(k)} f^{(k)}(-1) + w_1^{(k)} f^{(k)}(1)]$$

We have

$$E_{2p} = \int_{-1}^1 (x^2 - 1)^p dx = (-1)^p 2 \int_0^1 (1 - x^2)^p dx$$

Setting $x = \sin t$,

$$\begin{aligned} E_{2p} &= (-1)^p 2 \int_0^{\pi/2} \cos^{2p+1} t dt = (-1)^p 2 \frac{(2p)(2p-2) \cdots 2}{(2p+1)(2p-1) \cdots 3} \\ &= \frac{(-1)^p 2^{2p+1} (p!)^2}{(2p+1)!} \approx \frac{(-1)^p \sqrt{\pi p}}{p + 1/2} \end{aligned}$$

In particular

$$\begin{aligned} E_2 &= -4/3 \\ E_4 &= 16/15 \\ E_6 &= -25/35 \\ E_8 &= 256/315 \\ E_{10} &= -512/693 \end{aligned}$$

The third family is

$$\int_{-1}^1 f(x) dx = \sum_{k=0}^N [w_{-1}^{(k)} f^{(k)}(-1) + w_0^{(k)} f^{(k)}(0) + w_1^{(k)} f^{(k)}(1)]$$

with

$$E_{3p} = \begin{cases} \int_{-1}^1 x^p (x^2 - 1)^p dx & p \text{ even} \\ \int_{-1}^1 x^{p+1} (x^2 - 1)^p dx & p \text{ odd} \end{cases}$$

Setting $t = x^2$, we get

$$E_{3p} = \begin{cases} (-1)^p \int_0^1 t^{(p-1)/2} (1-t)^p dt & \text{even} \\ (-1)^p \int_0^1 t^{p/2} (1-t)^p dt & p \text{ odd} \end{cases}$$

These are beta functions whose values are

$$E_{3p} = \begin{cases} \frac{\Gamma[(p+1)/2] \Gamma(p+1)}{\Gamma[3(p+1)/2]} & p \text{ even} \\ \frac{\Gamma[(p+2)/2] \Gamma(p+1)}{\Gamma[(3p+4)/2]} & p \text{ odd} \end{cases}$$

Table 17.8*

Family	E_2	E_4	E_6	E_8	E_{10}	E_{12}	E_{14}	E_{16}
1-point	0.666667	0.400000	0.285714	0.222222	0.181818	0.153846	0.133333	0.117647
2-point	-1.333333	1.066667	-0.914286	0.812698	0.738817	0.681985	-0.636519	0.589077
3-point		0.266667	0.152381		-0.027706	0.017050		-0.003431
4-point		-0.118519		-0.030100		-0.003929		0.000796
5-point			-0.047619		0.007215			0.000468
6-point			-0.026819			0.002017		
7-point				-0.013169			0.000460	0.0002000
8-point				-0.008075				
9-point					-0.004380			
10-point					-0.002806			
11-point					-0.001616			
12-point						-0.001633		

* Computed by Mrs. Anne Duke.

Table 17.8 gives more results. The columns contain the values of E_m , and hence the error is of the form

$$\frac{E_m f^{(m)}(\theta)}{m!}$$

provided $G(s)$ is of constant sign.

We see that as we go up a column, the E_m tend to get larger, meaning that using more derivatives is not as good as using more points insofar as the size of the error coefficient is concerned; it still may be worth the machine time saved.

It is important to realize that Table 17.8 refers to one type of formula and represents an isolated experiment, not a definitive conclusion.

PROBLEMS

- 17.8.1 Check $G(s)$ for the three-point formulas:
 (a) function value only, E_4
 (b) function value plus first derivative, E_6 .
- 17.8.2 Discuss other reasonable experiments that might be tried.
- 17.8.3 Derive the four-point integration formulas for E_4 and E_8 .
- 17.8.4 (Harder) Write an essay on the effect on the weight used of the function value at that point when more derivatives are given there.

18

FORMULAS USING DIFFERENCES

18.1 USE OF DIFFERENCES

Before the age of large-scale digital computers most hand calculations were carried out in terms of function values and their differences. There were at least three reasons for this. First, differences generally use fewer digits than function values do—and so with hand calculations the arithmetic is that much easier. Second, the differences give immediate clues to the values of the derivatives, which are generally unobtainable. In particular we have the obvious approximations

$$\frac{\Delta y_n}{\Delta x} \approx \left. \frac{dy}{dx} \right|_{x=x_n+1/2}$$
$$\frac{\Delta^2 y_n}{\Delta x^2} \approx \left. \frac{d^2 y}{dx^2} \right|_{x=x_n+1}$$

Third, the difference table also gives a clue to the roundoff in the function values.

When computing machines came into widespread use, it became fairly clear that the first reason, fewer digits leading to faster arithmetic, was not very relevant, and the time to find the differences in a machine greatly exceeds any possible machine savings. Under the second reason, we still use differences as estimates

of derivatives, though to some extent this has fallen into disuse because the differences are often not found. The third reason of roundoff estimation was ignored chiefly because it soon became evident that the hand-computing experts had only an intuitive idea of the connection between roundoff and the differences. We have had to gradually work out a more explicit theory as given in Chap. 10. In particular, we had to quantify their impression of sign changes in the higher differences by first computing the correlation coefficient (Sec. 10.5) together with the corresponding probabilities of observing sign changes. We also had to work out the relation of the noise in the higher differences to the noise in the original function values (Sec. 10.4). This theory was intuitively understood, but the hand-computing experts could not say, with scientific detail, what they did so that it could be put on a computer. Indeed, they constantly maintained that it was an art and that it could not be quantified. Needless to say, in the theory presented in Chap. 10 some of their art was indeed lost, but some of it has been converted from intuition to measurable effects.

Today we find that much of current computing still ignores the approach through differences—the feeling persists that the differences take too much time and space in the computer and are not worth the trouble. The author believes, however, that going back to using them is a good idea and that many of the currently popular algorithms are poor substitutes for the direct use of the differences. For example, Romberg integration¹ effectively builds a table of differences of the answers at different spacings and tries to use this table to estimate the error. But the error comes from two distinct sources, the truncation error and the roundoff error. Taking more points in an integrand reduces the truncation error but builds up the roundoff, and in the behavior of the Romberg process it is hard to separate the two effects. With the direct use of the differences of the integrand values, the corresponding difference table enables us to separate the two effects and to make an intelligent guess about which is currently dominating and which strategy to adopt to improve the result—more function values may only increase the roundoff error!

18.2 NEWTON'S INTERPOLATION FORMULA

The first two methods for finding an interpolating polynomial through $N + 1$ points (the Vandermonde and Lagrange approaches) tacitly assumed that the number of points to be used was known. Often what is known is the accuracy desired, and the number of points to be used is determined as information about

¹ The purpose of Romberg integration is to eliminate the low-order-error terms of the error expansion in powers of Δx . The basic technique is usually attributed to Richardson.

the function is computed. Newton's interpolation formula, which we will now develop, is simply another way of writing the interpolating polynomial. It is useful because the number of points being used can easily be increased or decreased without repeating all the computation. The Newton formula of Sec. 9.6 is really the special case of equally spaced sample points.

As before, let the polynomial passing through the $N + 1$ points

$$(x_i, y_i) \quad (i = 1, 2, \dots, N + 1)$$

be labeled $P_N = P_N(x)$. We can write

$$P_N(x) = y_1 + (x - x_1)P_{N-1}(x)$$

where $P_{N-1}(x)$ is some polynomial of degree $N - 1$. It is clear that $P_N(x_1) = y_1$, and we are reduced to taking care of only N points ($i = 2, 3, \dots, N + 1$). The above equation can be written as

$$P_{N-1}(x) = \frac{P_N(x) - y_1}{x - x_1}$$

and hence we want

$$P_{N-1}(x_i) = \frac{P_N(x_i) - P_N(x_1)}{x_i - x_1} = \frac{y_i - y_1}{x_i - x_1} \quad i = 2, 3, \dots, N + 1$$

That is to say, we want $P_{N-1}(x)$ to pass through the points

$$\left(x_i, \frac{y_i - y_1}{x_i - x_1}\right) \quad i = 2, 3, \dots, N + 1$$

The quantities

$$\frac{y_i - y_1}{x_i - x_1} \equiv [x_i, x_1] \equiv [x_1, x_i]$$

are called "divided differences" and are customarily written with brackets.¹

The next step is, of course, to write

$$P_{N-1}(x) = [x_1, x_2] + (x - x_2)P_{N-2}(x)$$

and require $P_{N-2}(x)$ to take on the values of the divided differences of the divided differences,

$$[[x_i, x_1], [x_2, x_1]] = \frac{[x_i, x_1] - [x_2, x_1]}{x_i - x_2} = [x_i, x_2, x_1]$$

¹ Many other notations are used, for example $f[x_i, x_1]$ and $\rho[x_i, x_1]$.

It was easy to see that the first-order divided differences were independent of the order of the arguments in the brackets. We now show that the same is true of the second-order divided differences. If we start with three points

$$(x_1, y_1) \quad (x_2, y_2) \quad (x_3, y_3)$$

we get a unique polynomial (quadratic) through the three points. This may be written

$$y = y_1 + (x - x_1)\{[x_2, x_1] + (x - x_2)[x_3, x_2, x_1]\}$$

If we now choose the points in the order x_a, x_b , and x_c , we have

$$y = y_a + (x - x_a)\{[x_b, x_a] + (x - x_b)[x_c, x_b, x_a]\}$$

Since these are both the same quadratic, the coefficients of the x^2 term must be the same; that is, the two symbols

$$[x_3, x_2, x_1] = [x_c, x_b, x_a]$$

are in fact the same thing in alternative forms.

In general, we define

$$[x_1, x_2, x_3, \dots, x_n] \equiv \frac{[x_1, x_2, \dots, x_{n-1}] - [x_1, x_2, \dots, x_{n-2}, x_n]}{x_{n-1} - x_n}$$

and in exactly the same manner show that it is independent of the order of the x_i . Note that the denominator is the difference of the nonrepeated x 's in each $[\dots]$ taken in the same order.

One way to make a table of the needed values is as follows:

x_1	y^*			
		$[x_2, x_1]^*$		
x_2	y_2		$[x_3, x_2, x_1]^*$	
		$[x_3, x_1]$		$[x_4, x_3, x_2, x_1]^*$
x_3	y_3		$[x_4, x_2, x_1]$	.
		$[x_4, x_1]$	.	
x_4	y_4	.		.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

The asterisks indicate entries used as pivots in calculating succeeding values.

From this table we can write

$$y(x) = y_1 + (x - x_1)\{[x_2, x_1] + (x - x_2)\{[x_3, x_2, x_1] + (x - x_3)\{\dots\}\}\}$$

As a specific example, consider again the log table

x	$\log x$	[.]	[.,]	[.,.,]
1	0.0000*			
2	0.3010	0.30100*		
3	0.4771	0.23855	-0.06245*	
4	0.6021	0.20070	-0.05015	+0.01230*

The asterisks indicate the pivotal values.

Hence we get

$$y(x) = 0 + (x-1)\{0.3010 + (x-2)[(-0.06245) + (x-3)(0.01230)]\}$$

In particular,

$$\begin{aligned} y(2.5) &= 3/2\{0.3010 + 1/2[(-0.06245) + (-1/2)(0.01230)]\} \\ &= 0.40001 \end{aligned}$$

The correct value for $\log 2.5$ is 0.3979

PROBLEMS

- 18.2.1 Given a divided difference table, show how to add one more data point at the bottom of the table. Also show how to add one at the top. In the first case how does one change the Newton polynomial?
- 18.2.2 Make the divided difference table for $\Gamma(x)$ in Sec. 10.6.
- 18.2.3 Make the divided difference table for $\zeta(x)$ in Sec. 10.6.
- 18.2.4 Make the divided difference table for the error integral in Sec. 10.6.

18.3 AN ALTERNATIVE FORM FOR THE DIVIDED DIFFERENCE TABLE

The divided difference table, which lies at the heart of Newton's interpolation formula, may be written in an alternative and sometimes more useful form. This is based on the observation that

$$\begin{aligned} [x_1, x_2, \dots, x_n] &= \frac{[x_1, x_2, \dots, x_{n-1}] - [x_1, x_2, \dots, x_{n-2}, x_n]}{x_{n-1} - x_n} \\ &= \frac{[x_1, x_2, \dots, x_{n-1}] - [x_2, x_3, \dots, x_n]}{x_1 - x_n} \end{aligned}$$

In particular,

$$\begin{aligned}[x_1, x_2, x_3] &= \frac{[x_1, x_2] - [x_1, x_3]}{x_2 - x_3} \\ &= \frac{[x_1, x_2] - [x_2, x_3]}{x_1 - x_3}\end{aligned}$$

Thus we can write

x	y	[,]	[,,]	[,,,]
x_1	y_1			
		$[x_2, x_1]$		
x_2	y_2		$[x_3, x_2, x_1]$	
		$[x_3, x_2]$		$[x_4, x_3, x_2, x_1]$
x_3	y_3		$[x_4, x_3, x_2]$	
		$[x_4, x_3]$		
x_4	y_4			

While the entries in this table differ from those in the previous section, the entries in the top row, which are the ones used in Newton's formula, are the same.

Using the sample example of $\log x$, we get the following table:

x	$\log x$	[,]	[,,]	[,,,]
1	0.0000			
		0.3010		
2	0.3010		-0.06245	
		0.1761		+0.01230
3	0.4771		-0.02555	
		0.1250		
4	0.6021			

In this form, we may easily add a line at either end of the table and still expect the entries in the body of the table to vary smoothly. While there is no theoretical need to choose the points in numerical order, the smoothness in the table is lost if they are not. The first divided differences are secant lines and hence approximations to the first derivative in the interval (x_i, x_{i+1}) . Similarly, the second divided differences are local approximations to the second derivatives, etc.

PROBLEMS

18.3.1 Using the table

x	2	0	3	1
	8	0	27	1

compute the Newton interpolating polynomial. Show that $y = x^3$.

- 18.3.2 Compute the difference table both ways (Secs. 18.2 and 18.3) for $y = \sin x$ at a spacing of 30° :

x	$\sin x$
0°	0.0000
30°	0.5000
60°	0.8660
90°	1.0000

Find the approximating polynomial.

- 18.3.3 If you have only the top row of the divided difference table in storage in a computer, how can you add one point at the bottom of the table and compute the next divided difference? How do you add one point at the top and find the new line?

18.4 NEWTON'S FORMULA AT EQUAL SPACES

Very often our information about a function (in the form of samples) is given at a set of equally spaced values of x . This simplifies much of the notation and computation as well as the ideas involved.

For equally spaced data it is customary to use the notation of Part 1:

$$\Delta y_n = y_{n+1} - y_n$$

This is the familiar notation of the difference calculus except that we have fixed $\Delta x = h$ at all times:

$$\Delta x_n = x_{n+1} - x_n = h$$

We also have

$$\Delta^2 y_n = y_{n+2} - 2y_{n+1} + y_n$$

etc.

These differences are related to the divided differences as follows:

$$\begin{aligned} [x_2, x_1] &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y_1}{h} \\ [x_3, x_2, x_1] &= \frac{[x_3, x_2] - [x_2, x_1]}{x_3 - x_1} = \frac{\Delta y_2/h - \Delta y_1/h}{2h} \\ &= \frac{\Delta^2 y_1}{2! h^2} \end{aligned}$$

and, in general,

$$[x_1, x_2, \dots, x_n] = \frac{\Delta^{n-1} y_1}{(n-1)! h^{n-1}}$$

The differences are approximations to derivatives at the middle of the range of the samples used:

$$\begin{aligned}\Delta y_1 &\approx h \frac{dy(x+h/2)}{dx} \\ \Delta^2 y_1 &\approx h^2 \frac{d^2 y(x+h)}{dx^2} \\ &\dots\dots\dots \\ \Delta^n y_1 &\approx h^n \frac{d^n y(x+nh/2)}{dx^n}\end{aligned}$$

It is these relationships that enable the finite difference calculus to approximate expressions in the differential calculus.

Newton's formula in this new notation for equally spaced sample points is (Secs. 9.6 and 18.2)

$$\begin{aligned}y = y_0 + (x - x_0) \frac{\Delta y_0}{h} + (x - x_0)(x - x_0 - h) \frac{\Delta^2 y_0}{2h^2} \\ + (x - x_0)(x - x_0 - h)(x - x_0 - 2h) \frac{\Delta^3 y_0}{3!h^3} + \dots\end{aligned}$$

If we suppose that $x_0 = 0$, we get

$$y = y_0 + x \frac{\Delta y_0}{h} + x(x - h) \frac{\Delta^2 y_0}{2h^2} + x(x - h)(x - 2h) \frac{\Delta^3 y_0}{3!h^3} + \dots$$

and if we further assume $h = 1$, we have

$$y = y_0 + x \Delta y_0 + x(x - 1) \frac{\Delta^2 y_0}{2!} + x(x - 1)(x - 2) \frac{\Delta^3 y_0}{3!} + \dots$$

PROBLEMS

18.4.1 Write the error term for Newton's formula when the data are given at $-n, -(n-1), \dots, 0, 1, 2, \dots, n$ and a $(2n)$ th difference is used in the formula.

18.5 INTERPOLATION IN TABLES

One of the major uses of equally spaced interpolation formulas is for interpolation in tables of equally spaced data. A typical example is the error integral whose difference table is given in Sec. 10.6. The differences are well behaved in the sense

that they first tend toward zero, being fairly smooth in the third-difference column, but beginning with the fourth difference, and clearly in the fifth difference where the sign is almost exactly alternating, we find that roundoff noise is dominating. For this table we would not expect, then, to go beyond fourth differences in any interpolation formula.

In principle, Newton's formula for interpolation gives the answer, and except for roundoff effects the answer is exactly the same as could be found by any other interpolating polynomial using the same sample points. Nevertheless, in practice, many other formulas are used, and in the interest of general education, so that the reader can understand the methods given in the prefaces of most tables, we shall discuss some of those which are most popular.

18.6 THE LOZENGE DIAGRAM

The lozenge diagram is a device for showing that a large number of formulas which appear to be different are really all the same. We have used the notation for the binomial coefficients

$$C(u+k, n) = \frac{(u+k)(u+k-1)(u+k-2) \cdots (u+k-n+1)}{n!}$$

There are n factors in the numerator and n in the denominator. Viewed as a function of u , $C(u+k, n)$ is a polynomial of degree n .

Figure 18.6.1 shows the lozenge diagram. A line starting at a point on the left edge and following some path across the page defines an interpolation formula if the following rules are used.

- 1a For a *left-to-right* step, *add*.
- 1b For a *right-to-left* step, *subtract*.
- 2a If the *slope* of the step is *positive*, use the product of the difference crossed times the factor immediately *below*.
- 2b If the *slope* of the step is *negative*, use the product of the difference crossed times the factor immediately *above*.
- 3a If the step is *horizontal* and passes through a *difference*, use the product of the difference times the *average* of the factors *above and below*.
- 3b If the step is *horizontal* and passes through a *factor*, use the product of the factor times the *average* of the differences *above and below*.

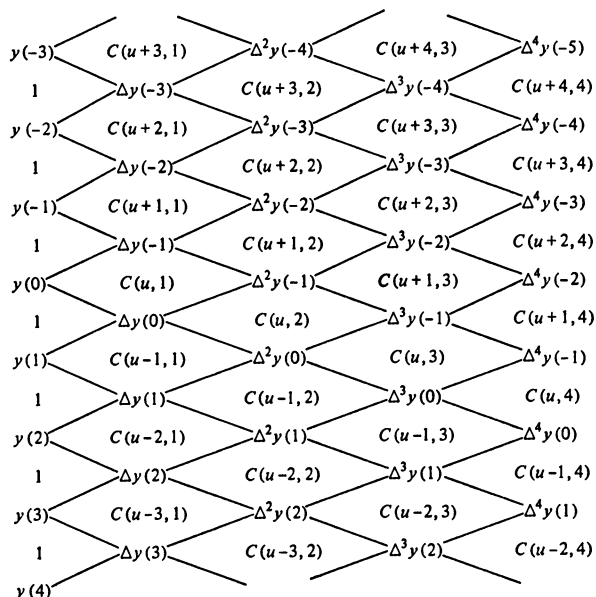


FIGURE 18.6.1

As an example of rules 1a and 2b, consider starting at $y(0)$ and going down to the right. We get, term by term,

$$\begin{aligned} y(u) &= y(0) + C(u, 1) \Delta y(0) + C(u, 2) \Delta^2 y(0) + C(u, 3) \Delta^3 y(0) + \cdots \\ &= y(0) + u \Delta y(0) + \frac{u(u-1)}{2} \Delta^2 y(0) + \frac{u(u-1)(u-2)}{3!} \Delta^3 y(0) + \cdots \end{aligned}$$

which is Newton's formula.

Had we gone up and to the right, we would have used rules 1a and 2a to get Newton's backward formula:

$$\begin{aligned} y(u) &= y(0) + C(u, 1) \Delta y(-1) + C(u+1, 2) \Delta^2 y(-2) \\ &\quad + C(u+2, 3) \Delta^3 y(-3) + \cdots \\ &= y(0) + u \Delta y(-1) + \frac{(u+1)u}{2} \Delta^2 y(-2) + \frac{(u+2)(u+1)u}{3!} \Delta^3 y(-3) + \cdots \end{aligned} \tag{18.6.1}$$

To get Stirling's formula, we start at $y(0)$ and go horizontally to the right, using rules 3a and 3b:

$$\begin{aligned}
 y(u) &= y(0) + u \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{C(u+1, 2) + C(u, 2)}{2} \Delta^2 y_{-1} \\
 &\quad + C(u+1, 3) \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \dots \\
 &= y_0 + u \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{u^2}{2} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \dots
 \end{aligned} \tag{18.6.2}$$

If we start midway between $y(0)$ and $y(1)$, we get Bessel's formula:

$$\begin{aligned}
 y(u) &= 1 \frac{y_0 + y_1}{2} + \frac{C(u, 1) + C(u-1, 1)}{2} \Delta y_0 + C(u, 2) \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \dots \\
 &= \frac{y_0 + y_1}{2} + (u - \tfrac{1}{2}) \Delta y_0 + \frac{u(u-1)}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \dots
 \end{aligned} \tag{18.6.3}$$

If we zigzag properly, we can get Gauss' formula for interpolation:

$$y(u) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2} \Delta^2 y(-1) + \frac{u(u^2-1)}{3!} \Delta^3 y(-1) + \dots \tag{18.6.4}$$

All sorts of paths can be chosen, and each will give some formula. What we need to show is that these are all valid interpolation formulas. Such a proof requires showing the following:

- 1 At least one valid formula results, and since we have found Newton's interpolation formula from the diagram, this step is completed.
- 2 The contribution around any closed path is zero; hence we may deform a path into any other path that we wish.
- 3 If two formulas end at the same place, they are the same. It is necessary to prove this fact since the entry points into the lozenge diagram need not be the same for different formulas.

To prove step 2, we take a single complete lozenge (Fig. 18.6.2).

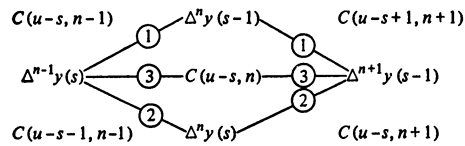


FIGURE 18.6.2

$$\text{Path 1: } C(u-s, n) \Delta^n y(s-1) + C(u-s+1, n+1) \Delta^{n+1} y(s-1)$$

$$\text{Path 2: } C(u-s, n) \Delta^n y(s) + C(u-s, n+1) \Delta^{n+1} y(s-1)$$

$$\begin{aligned} \text{Path 3: } & C(u-s, n) \frac{\Delta^n y(s-1) + \Delta^n y(s)}{2} \\ & + \frac{C(u-s+1, n+1) + C(u-s, n+1)}{2} \Delta^{n+1} y(s-1) \end{aligned}$$

Recalling that going from right to left produces negative terms (rule 1b), we have to show only that the three paths are the same.

Path 1 – path 2

$$\begin{aligned} &= C(u-s, n) [\Delta^n y(s-1) - \Delta^n y(s)] \\ &\quad + [C(u-s+1, n+1) - C(u-s, n+1)] \Delta^{n+1} y(s-1) \\ &= C(u-s, n) [-\Delta^{n+1} y(s-1)] \\ &\quad + C(u-s, n) \frac{(u-s+1) - (u-s-n)}{n+1} \Delta^{n+1} y(s-1) \\ &= C(u-s, n) [-\Delta^{n+1} y(s-1) + \Delta^{n+1} y(s-1)] = 0 \end{aligned}$$

Also, path 3 equals the average of paths 1 and 2. Hence step 2 of the proof is complete.

To examine step 3 of the proof, we use Fig. 18.6.3.

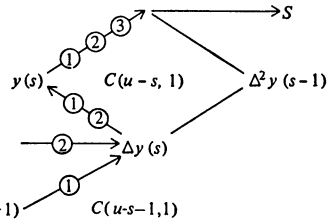


FIGURE 18.6.3

$$\text{Path 1} = y(s+1) + C(u-s-1, 1) \Delta y(s) - C(u-s, 1) \Delta y(s) + S$$

$$\begin{aligned} \text{Path 2} = & \frac{y(s+1) + y(s)}{2} + \frac{C(u-s-1, 1) + C(u-s, 1)}{2} \Delta y(s) \\ & - C(u-s, 1) \Delta y(s) + S \end{aligned}$$

$$\text{Path 3} = y(s) + S$$

where S is the rest of the formula. Now using

$$y(s+1) - y(s) = \Delta y(s)$$

we obtain

$$\begin{aligned} \text{Path 1} &= y(s) + \Delta y(s) + (u-s) \Delta y(s) - \Delta y(s) - (u-s) \Delta y(s) + S \\ &= y(s) + S = \text{path 3} \end{aligned}$$

Similarly, path 2 may be reduced to path 3. Thus we conclude that the interpolation formula depends on the ending values, not on the path used to reach them.

18.7 REMARKS ON THESE FORMULAS

How do these different formulas compare with each other and with the Lagrange formula found earlier? The value obtained in an interpolation depends on the polynomial used, and the polynomial depends on the sample points actually used (some formulas might appear to use values not needed). The error term has the form (see Sec. 14.6)

$$\frac{(x-x_1)(x-x_2)\cdots(x-x_{n+1})y^{(n+1)}(\bar{x})}{(n+1)!}$$

The coefficient of the derivative is minimized when x is in the middle of the range of samples. Thus there is a tendency to use an even number of samples when the interpolation point is in the middle of an interval and an odd number when it is near a sample point.

The methods derived by the lozenge diagram explicitly exhibit the differences, and these differences give some idea of the accuracy but require more arithmetic than does the Lagrange method of interpolating. The Lagrange method, however, does not directly use the differences, and so no indication of the accuracy is available. This is one of the fundamental problems in numerical computation—the use of differences gives some clues to the accuracy of the method being used, while the Lagrange approach minimizes the amount of arithmetic. As a result, the difference methods tend to be used in exploratory work, and the Lagrange methods in well-understood, routine work.

18.8 MISCELLANEOUS INTERPOLATION FORMULAS

A number of other formulas which are not directly obtained from the lozenge diagram are occasionally useful. They usually rest on the fact that any particular difference may be eliminated by using

$$\Delta^n y(s+1) - \Delta^n y(s) = \Delta^{n+1} y(s)$$

The cost of this elimination is the inclusion of an extra difference of some other order.

For example, in Bessel's interpolation formula we may eliminate all odd-order differences; this gives

$$y(u) = (1-u)y(0) + uy(1) + \frac{(2-u)(1-u)(-u)}{3!} \Delta^2 y(-1) \\ + \frac{(u+1)u(u-1)}{3!} \Delta^2 y(0) + \dots$$

which is known as Everett's formula and is quite popular since the table maker need only publish the function and even-order differences.

Similarly we may start with almost any formula and eliminate differences of any order that we please (at the cost of several differences of the orders that we leave in the formula). If we wish, we could eliminate, say, both the second- and third-order differences, thus using the function, the first differences, and the fourth and higher differences. Carried to the extreme, this leads to the Lagrange form which uses no differences but many function values. The purpose of eliminating the differences is mainly to save type and space in printing the table, and this is done at the cost of extra work on the part of the user. The proper balance depends on circumstances and cannot be given once and for all.

A second device often used is called "throwback." This idea, due mainly to Comrie, uses the fact that the coefficients of the successive differences tend to be proportional to each other in various interpolation formulas, such as Everett's; consequently, if suitable amounts of the higher differences are combined with those of the lower when the table is printed, then much of the effect of the higher-order differences in the interpolation is automatically achieved by using the lower-order formula.

Thus, in Bessel's formula, the ratio of the coefficients of the Δ^4 to the coefficient of Δ^2 is

$$\frac{B^{IV}}{B^{II}} = \frac{(u+1)(u-2)}{12} \quad 0 \leq u \leq 1$$

which varies from $-1/6$ to $-3/16$. Hence if we put c units of Δ^4 with Δ^2 to form the modified second difference column, we make an error of

$$(B^{IV} - cB^{II}) \Delta^4$$

The number c is often taken as -0.184 as an average compromise.

We again remind the reader that table making is a highly skilled art and refer the reader to Fox [12] as well as to the standard text of Ralston [49].

18.9 THE HAMMING-PINKHAM INTEGRATION FORMULA

We now turn to a family of integration formulas whose *correction terms* (truncation errors) are expressed as differences of the function values. The effect of taking more differences is to increase the accuracy of the formula as judged by the degree of the polynomial for which it is exactly true (within roundoff). Thus we can *sequentially* try successive members of the family to find the most appropriate one instead of more or less committing ourselves to the accuracy *before* starting (as we must with other types of formulas). Because we have the difference table of the integrand values we can make an estimate of the roundoff errors independently of the estimate of the truncation error.

These formulas are closely related to the composite integration formulas discussed in Chap. 20, and so it is worth a few moments to explain the idea of a composite formula. When we find that the formula such as Simpson's is not accurate enough, we usually decide to use more points; with more points we can make the formula more accurate in the sense of being exact for a higher order polynomial. Instead of using this method we can apply the same basic formula to a sequence of intervals and add the results of the individual integrals. This second approach leads to the idea of a *composite formula*. The two best-known examples are the composite trapezoid rule

$$\int_0^{nh} f(x) dx = h \left\{ \frac{1}{2} f(0) + f(h) + f(2h) + \cdots + f[(n-1)h] + \frac{1}{2} f(nh) \right\}$$

and the composite Simpson's formula

$$\int_0^{2nh} f(x) dx = \frac{h}{3} \{ f(0) + 4f(h) + 2f(2h) + 4f(3h) + \cdots + 4f[(2n-1)h] + f(2nh) \}$$

where in both cases we have used the basic integration formula repeatedly.

The classic formula which led to the investigation of this family of formulas is the Gregory¹ formula (1670)

$$\begin{aligned} \int_0^n f(x) dx &= \frac{1}{2} f(0) + f(1) + f(2) + \cdots + f(n-1) + \frac{1}{2} f(n) \\ &\quad - \frac{1}{12} [\Delta f(n-1) - \Delta f(0)] - \frac{1}{24} [\Delta^2 f(n-2) + \Delta^2 f(0)] \\ &\quad - \frac{19}{720} [\Delta^3 f(n-3) - \Delta^3 f(0)] - \frac{9}{480} [\Delta^4 f(n-4) + \Delta^4 f(0)] \\ &\quad - \frac{863}{60,480} [\Delta^5 f(n-5) - \Delta^5 f(0)] + \cdots \end{aligned}$$

¹ James Gregory (1638–1675).

This formula is clearly the composite trapezoid rule *with end corrections* that make the formula as accurate as the differences included indicate. The alternating sign between the differences has the effect of causing the accuracy to jump by 2 or 0 as one more term of differences is included; thus we tend to stop on an even-order difference and get the next odd-order one “free.”

A similar formula based on Simpson's composite formula is

$$\begin{aligned}\int_0^{2n} f(x) dx &= \frac{1}{3} f(0) + \frac{4}{3} f(1) + \frac{2}{3} f(2) + \cdots + \frac{4}{3} f(2n-1) + \frac{1}{3} f(2n) \\ &\quad - \frac{4}{720} [\Delta^3 f(2n-3) - \Delta^3 f(0)] - \frac{4}{480} [\Delta^4 f(2n-4) + \Delta^4 f(0)] \\ &\quad - \frac{548}{60,480} [\Delta^5 f(2n-5) - \Delta^5 f(0)] + \cdots\end{aligned}$$

where the coefficients of the differences are *all* smaller in size than those of the Gregory formula (see Table 18.10 for more coefficients).

A still more interesting formula is the Hamming-Pinkham formula

$$\begin{aligned}\int_0^{2n} f(x) dx &= 2[f(1) + f(3) + \cdots + f(2n-1)] \\ &\quad + \frac{1}{6} [\Delta f(2n-1) - \Delta f(0)] + \frac{1}{12} [\Delta^2 f(2n-2) + \Delta^2 f(0)] \\ &\quad + \frac{13}{360} [\Delta^3 f(2n-3) - \Delta^3 f(0)] + \frac{1}{80} [\Delta^4 f(2n-4) + \Delta^4 f(0)] \\ &\quad + \frac{41}{30,240} [\Delta^5 f(2n-5) - \Delta^5 f(0)] + \cdots\end{aligned}$$

where only *half* of the integrand values in the basic summation and comparatively few more at the ends are needed to form the differences. Thus the coarse sampling rate is used in the main part of the integrand, and double the rate is used at the ends to estimate the differences. This means that about half the computing is saved for the same order of accuracy.

The first two cases, the Gregory and Simpson composite formulas, can be used in an adaptive mode of computation by halving the spacing if the truncation error is too large; the third formula requires taking one-third of the spacing if the old integrand values are to be used in the next step.

In order to compare these three formulas (and some others) we shall, as

usual, consider a family of formulas which includes them as special cases. This family is

$$\int_0^{2n} f(x) dx = af(0) + bf(1) + 2af(2) + \cdots + af(2n) \\ + \sum_{s=0}^{\infty} q_s \{ \Delta^s f(2n-s) + (-1)^s \Delta^s f(0) \}$$

where we want both the coefficients a and b and the q_s to be independent of n , which is the number of double intervals used.

In the polynomial approach to finding formulas we require the formula to be exact for successive powers of x ; that is, we set down the defining equations

$$\int_0^{2n} x^p dx = a(0)^p + b(1)^p + 2a(2)^p + b(3)^p + \cdots + a(2n)^p \\ + \sum_{s=0}^{\infty} q_s \{ \Delta^s (2n-s)^p + (-1)^s \Delta^s (0)^p \} \quad p = 0, 1, 2, \dots$$

For $p = 0$ we have

$$2n \equiv nb + [2 + 2(n-1)]a = 2na + nb$$

or

$$b = 2(1-a)$$

which means we have a one parameter family of formulas depending on a .

18.10 THE DERIVATION OF THE FORMULAS

The derivation of the Hamming-Pinkham formula is based on the method of generating functions (Sec. 11.4). We multiply the p th defining equation by $t^p/p!$ and sum to get (using the power-series expansion of the exponential)

$$\int_0^{2n} e^{tx} dx = ae^0 + be^t + 2ae^{2t} + \cdots + ae^{2nt} \\ + \sum_{s=0}^{\infty} q_s (\Delta^s [e^{2n-s}]^t + (-1)^s \Delta^s [e^0]^t)$$

But noting that this is a composite formula with weights

$$\begin{array}{cccccccc} a & b & & a & & & & \\ & & a & b & a & & & \\ & & & & a & b & a & \\ & & & & & & a & \dots \\ \hline a & b & 2a & b & 2a & b & 2a & \dots \end{array}$$

we see that the first sum on the right-hand side is

$$(ae^{-t} + b + ae^t)(e^t + e^{3t} + \cdots + e^{(2n-1)t}) \\ = (ae^{-t} + b + ae^t) \frac{e^{2nt} - 1}{e^t - e^{-t}}$$

The differences that occur in the formula can be written as

$$\begin{aligned} \Delta^s(e^{2n-s}t) &= \Delta^{s-1}(e^{(2n-s+1)t} - e^{(2n-s)t}) = \Delta^{s-1}(e^{(2n-s+1)t})(1 - e^{-t}) \\ &\vdots \\ &= e^{2nt}(1 - e^{-t})^s \\ (-1)^s \Delta^s(e^{0t}) &= (-1)^s \Delta^{s-1}(e^t - e^0) = (-1)^s \Delta^{s-1}(e^t - 1) = (-1)^{s-1} \Delta^{s-1}(1 - e^t) \\ &\vdots \\ &= (1 - e^t)^s \end{aligned}$$

Performing the integration on the left-hand side of the formula and assembling the results, we have

$$\frac{e^{2nt} - 1}{t} = (ae^t + b + ae^{-t}) \frac{e^{2nt} - 1}{e^t - e^{-t}} + \sum_{s=0}^{\infty} q_s [e^{2nt}(1 - e^{-t})^s + (1 - e^t)^s]$$

If we now define $Q(t)$ as

$$Q(t) = \frac{1}{t} - \frac{ae^t + b + ae^{-t}}{e^t - e^{-t}} - \sum_{s=0}^{\infty} q_s (1 - e^{-t})^s$$

then we see that we can write the equation as

$$e^{2nt} Q(t) + Q(-t) = 0$$

This equation is true, if $Q(t) \equiv 0$. If $Q(t) = 0$, then

$$\sum_{s=0}^{\infty} q_s (1 - e^{-t})^s = \frac{1}{t} - \frac{ae^t + b + ae^{-t}}{e^t - e^{-t}}$$

and defines the q_s .

It is natural to set

$$1 - e^{-t} \equiv v \quad \text{or} \quad e^{-t} = 1 - v \quad t = -\ln(1 - v)$$

so that we have a power series in v with coefficients q_s . Using $b = 2(1 - a)$, we have

$$\begin{aligned} \sum_{s=0}^{\infty} q_s v^s &= \frac{-1}{\ln(1 - v)} - \frac{a/(1 - v) + 2 - 2a + a(1 - v)}{1/(1 - v) - (1 - v)} \\ &= \frac{-1}{\ln(1 - v)} - \frac{2 - 2v + av^2}{2v - v^2} \\ &= \frac{-1}{\ln(1 - v)} + a - \frac{1}{v} - \frac{2a - 1}{2 - v} \end{aligned}$$

The choice of $a = 1/2$ eliminates one term and gives the Gregory formula (as a check on the work so far). For $a = 1/2$ we have

$$\sum_{s=0}^{\infty} q_s v^s = \frac{-1}{\ln(1-v)} - \frac{1}{v} + \frac{1}{2}$$

to determine the q_s . Using

$$-\ln(1-v) = v + \frac{v^2}{2} + \frac{v^3}{3} + \frac{v^4}{4} + \cdots$$

and dividing out $1/\ln(1-v)$, we get

$$\sum q_s v^s = -\frac{v}{12} - \frac{v}{24} - \frac{19}{720} v^3 - \cdots$$

which checks with Gregory's formula.

If we label the coefficients of Gregory's formula g_s , we have for the general case

$$\begin{aligned} \sum_{s=0}^{\infty} q_s v^s &= \frac{-1}{\ln(1-v)} + a - \frac{1}{v} - \frac{2a-1}{2-v} \\ &= \sum_{s=0}^{\infty} g_s v^s + (a - \tfrac{1}{2}) - \frac{2a-1}{2-v} \\ &= \sum_{s=0}^{\infty} g_s v^s - (a - \tfrac{1}{2}) \frac{v}{2-v} \end{aligned}$$

Dividing out $v/(2-v)$, we get

$$\sum_{s=0}^{\infty} q_s v^s = \sum_{s=0}^{\infty} g_s v^s - (a - \tfrac{1}{2}) \sum_{s=1}^{\infty} \left(\frac{v}{2}\right)^s$$

Equating coefficients of v^s , we have $q_0 = g_0$ and for $s = 0$

$$q_s = g_s - (a - \tfrac{1}{2}) 2^{-s}$$

From this, Table 18.10 is easily compiled and is displayed in Fig. 18.10.1.

A remarkable formula results by setting $a = 0$ ($b = 2$) since now half the integrand values have weight zero and so do not enter into the summation.

$$\begin{aligned} \int_0^{2n} f(x) dx &= 2 \sum_{r=0}^{n-1} f(2r+1) + \frac{1}{6} [\Delta f(2n-1) - \Delta f(0)] \\ &+ \frac{1}{12} [\Delta^2 f(2n-2) + \Delta^2 f(0)] + \frac{13}{360} [\Delta^3 f(2n-3) - \Delta^3 f(0)] \\ &+ \frac{1}{80} [\Delta^4 f(2n-4) + \Delta^4 f(0)] + \frac{41}{30,240} [\Delta^5 f(2n-5) - \Delta^5 f(0)] + \cdots \end{aligned}$$

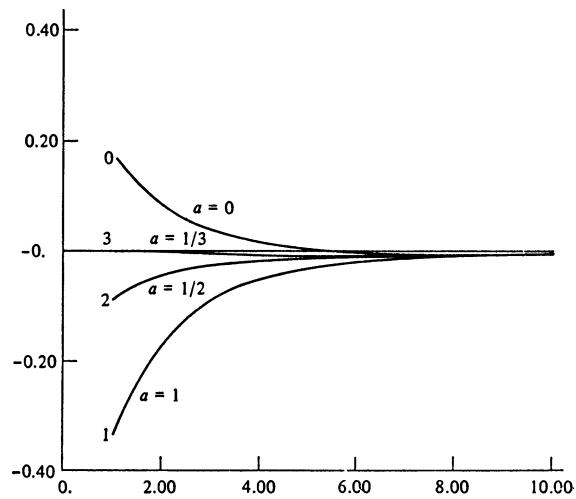


FIGURE 18.10.1

Table 18.10

s	Gregory $a = 1/2$	Simpson $a = 1/3$	$a = 0$	$a = 1$
1	$-\frac{1}{12}$	0	$\frac{2}{12}$	$-\frac{4}{12}$
2	$-\frac{1}{24}$	0	$\frac{2}{24}$	$-\frac{4}{24}$
3	$-\frac{19}{720}$	$-\frac{4}{720}$	$\frac{26}{720}$	$-\frac{64}{720}$
4	$-\frac{9}{480}$	$-\frac{4}{480}$	$\frac{6}{480}$	$-\frac{24}{480}$
5	$-\frac{863}{60480}$	$-\frac{548}{60480}$	$\frac{82}{60480}$	$-\frac{1808}{60480}$
6	$-\frac{275}{24192}$	$-\frac{212}{24192}$	$-\frac{86}{24192}$	$-\frac{464}{24192}$
7	$-\frac{33953}{3628800}$	$-\frac{29228}{3628800}$	$-\frac{19778}{3628800}$	$-\frac{48128}{3628800}$
8	$-\frac{8183}{1036800}$	$-\frac{7508}{1026800}$	$-\frac{6158}{1036800}$	$-\frac{10208}{1036800}$
9	$-\frac{3250433}{479001600}$	$-\frac{3094508}{479001600}$	$-\frac{2782658}{479001600}$	$-\frac{3718208}{479001600}$
10	$-\frac{14013}{2365440}$	$-\frac{13628}{2365440}$	$-\frac{12858}{2365440}$	$-\frac{15168}{2365440}$

Experimental results show that this formula compares favorably with other members of the family; indeed the results are somewhat better on the average. We can, using this formula, estimate the truncation error made by dropping the differences past some order k , and we can estimate the roundoff from the difference table.

PROBLEMS

18.10.1 Derive Gregory's formula when n is an odd number.

18.10.2 Consider using five points (four intervals) in the basic formula and derive a two-parameter family of formulas.

18.10.3 Derive the midpoint formula (Bickley's formula). See [8].

$$\begin{aligned} \int_0^n f(x) dx = \sum_{k=0}^{n-1} f\left(k + \frac{1}{2}\right) &+ \frac{1}{24} [\Delta f_{n-3/2} - \Delta f_{1/2}] + \frac{1}{24} [\Delta^2 f_{n-5/2} + \Delta^2 f_{1/2}] \\ &+ \frac{223}{5,760} [\Delta^3 f_{n-7/2} - \Delta^3 f_{1/2}] + \frac{103}{2,880} [\Delta^4 f_{n-9/2} + \Delta^4 f_{1/2}] \\ &+ \frac{32,119}{967,680} [\Delta^5 f_{n-11/2} - \Delta^5 f_{1/2}] + \cdots \end{aligned}$$

18.10.4 Derive

$$\int_0^1 f(x) dx = \sum_{k=0}^{\infty} A_k [f^{2k}(0) + f^{2k}(1)]$$

where

$$A_{k-1} = \frac{2(2^{2k} - 1)}{(2k)!} B_{2k}$$

*FORMULAS USING THE SAMPLE POINTS AS PARAMETERS

19.1 INTRODUCTION

Up to now we have been using the weights in the formulas as the parameters. Since the weights occurred linearly in the defining equations, they were easy to find (the determinant of the system was shown to be nonzero, and we could use the sampling polynomials to find the w_i directly if we wanted to). We now consider using the position x_i of the samples as *additional* parameters to make the formulas exact for more powers of x . The new parameters do *not* occur linearly, and so the defining equations will be harder to solve. But again it should be pointed out that the defining equations are solved only once when the formula is being developed, not each time the formula is used. The gain in making the formula exact for more powers of x while using *the same number* of sample points is obvious.

The resulting formulas generally have awkward values for the sample positions x_i and the weights w_i , and they were therefore neglected in hand-calculating days. This is no reason to neglect them when computing machines are readily available. The principal remaining objection is that the peculiar values of the x_i and w_i must be provided to the routine by keypunching and storing them

from published tables¹ or from other routines which found them and saved the results.

The mathematical theory is both elegant and extensive, and we will give only a minimum of material to indicate what is possible.

19.2 SOME EXAMPLES

Before plunging into the theory associated with the general case, we shall examine a number of special cases to orient ourselves. First consider estimating the following integral from two samples:

$$\int_0^{\infty} e^{-x} f(x) dx = w_1 f(x_1) + w_2 f(x_2)$$

where the parameters are w_1 , w_2 , x_1 , and x_2 . The four defining equations are, using the first four powers of x ,

$$\begin{array}{lcl} 1: & m_0 = 1 = w_1 & + w_2 \\ x: & m_1 = 1 = w_1 x_1 & + w_2 x_2 \\ x^2: & m_2 = 2 = w_1 x_1^2 & + w_2 x_2^2 \\ x^3: & m_3 = 6 = w_1 x_1^3 & + w_2 x_2^3 \end{array} \left| \begin{array}{c} c_0 \\ c_1 \\ 1 \\ 1 \end{array} \right| \begin{array}{c} c_0 \\ c_1 \\ c_1 \\ 1 \end{array}$$

We now use the polynomial

$$\pi(x) = (x - x_1)(x - x_2) \equiv x^2 + c_1 x + c_0$$

and multiply the top equation by c_0 , the second equation by c_1 , and the third equation by 1, and then add the three equations. We get

$$\begin{aligned} 2 + c_1 + c_0 &= w_1(x_1^2 + c_1 x_1 + c_0) + w_2(x_2^2 + c_1 x_2 + c_0) \\ &= w_1 \pi(x_1) + w_2 \pi(x_2) = 0 \end{aligned}$$

since $\pi(x_i) = 0$. Shifting the three multipliers down one equation and repeating the process, we get

$$6 + 2c_1 + c_0 = w_1 x_1 \pi(x_1) + w_2 x_2 \pi(x_2) = 0$$

Thus we have the pair of equations

$$\begin{aligned} c_1 + c_0 &= -2 \\ 2c_1 + c_0 &= -6 \end{aligned}$$

¹ For $\int_{-1}^1 f(x) dx$, $\int_0^{\infty} e^{-x} f(x) dx$, $\int_{-\infty}^{\infty} e^{-x^2} f(x) dx$ and many others, see National Bureau of Standards, "Handbook of Mathematical Functions," and Dover Publications, Inc., New York, 1964. Also see Mineur [41] and Stroud and Secrest [57].

whose solution is

$$\begin{aligned}c_1 &= -4 \\c_0 &= 2\end{aligned}$$

Since these are the coefficients of $\pi(x)$, we have

$$\pi(x) = x^2 - 4x + 2 = 0$$

and the sample points are

$$\begin{aligned}x_1 &= 2 - \sqrt{2} \\x_2 &= 2 + \sqrt{2}\end{aligned}$$

We are now reduced to the case of Chap. 15 with known sample points. The first two defining equations are:

$$\begin{aligned}1 &= w_1 + w_2 \\1 &= (2 - \sqrt{2})w_1 + (2 + \sqrt{2})w_2\end{aligned}$$

whose solutions are:

$$\begin{aligned}w_1 &= \frac{\sqrt{2} + 1}{2\sqrt{2}} \\w_2 &= \frac{\sqrt{2} - 1}{2\sqrt{2}}\end{aligned}$$

Our formula is, therefore,

$$\int_0^\infty e^{-x} f(x) dx = \frac{\sqrt{2} + 1}{2\sqrt{2}} f(2 - \sqrt{2}) + \frac{\sqrt{2} - 1}{2\sqrt{2}} f(2 + \sqrt{2})$$

which is exact for cubics using only two samples of the integrand.

How did we know that the weights we found from the first two defining equations would satisfy the last two equations? The answer is simple; we chose the x_i so that the last two equations were linear combinations (with coefficients c_0 , c_1 , and 1) of the first two equations; hence the w_i automatically satisfied the last two equations.

As a second example, consider

$$\int_{-1}^1 f(x) dx = w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)$$

With six parameters we can write six defining equations:

$$\begin{array}{rcl}
 2 & = & w_1 + w_2 + w_3 \\
 0 & = & w_1 x_1 + w_2 x_2 + w_3 x_3 \\
 2/3 & = & w_1 x_1^2 + w_2 x_2^2 + w_3 x_3^2 \\
 0 & = & w_1 x_1^3 + w_2 x_2^3 + w_3 x_3^3 \\
 2/5 & = & w_1 x_1^4 + w_2 x_2^4 + w_3 x_3^4 \\
 0 & = & w_1 x_1^5 + w_2 x_2^5 + w_3 x_3^5
 \end{array}
 \begin{array}{c}
 \left| \begin{array}{c} c_0 \\ c_1 \\ c_2 \\ 1 \\ 1 \\ 1 \end{array} \right| \\
 \left| \begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_1 \\ c_2 \\ 1 \end{array} \right| \\
 \left| \begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_1 \\ c_2 \\ 1 \end{array} \right|
 \end{array}$$

We again use the appropriate $\pi(x)$,

$$\pi(x) \equiv (x - x_1)(x - x_2)(x - x_3) \equiv x^3 + c_2 x^2 + c_1 x + c_0$$

By multiplying by the c_1 as shown on the right of the defining equations and adding the results, we get three equations for c_0 , c_1 , and c_2 :

$$2c_0 + 0c_1 + \frac{2}{3}c_2 + 0 \cdot 1 = 0$$

$$0c_0 + \frac{2}{3}c_1 + 0c_2 + \frac{2}{5} \cdot 1 = 0$$

$$\frac{2}{3}c_0 + 0c_1 + \frac{2}{5}c_2 + 0 \cdot 1 = 0$$

Clearly $c_0 = c_2 = 0$ and $c_1 = -3/5$. Hence

$$\pi(x) = x^3 + 0x^2 - \frac{3}{5}x + 0 = 0$$

$$x_1 = -\sqrt{3/5}$$

$$x_2 = 0$$

$$x_3 = \sqrt{3/5}$$

Knowing the sample points x_i , we have the defining equations

$$2 = w_1 + w_2 + w_3$$

$$0 = -\sqrt{3/5} w_1 + \sqrt{3/5} w_3$$

$$2/3 = \frac{2}{3} w_1 + \frac{2}{3} w_3$$

It follows from the second that $w_1 = w_3$, from the third equation that

$$w_1 = w_3 = 5/9$$

and finally from the first equation $w_2 = 8/9$. The formula is, therefore,

$$\int_{-1}^1 f(x) dx = \frac{5}{9} f(-\sqrt{3/5}) + \frac{8}{9} f(0) + \frac{5}{9} f(\sqrt{3/5})$$

and is exact for fifth-degree polynomials (see Sec. 16.9).

If we had used the obvious symmetry, we would have realized that $w_1 = w_3$, $x_1 = -x_3$, and $x_2 = 0$ would make the formula true for *all* odd powers of x . Hence we would have had the defining equations

$$\begin{aligned} 1: \quad 2 &= 2w_1 + w_2 \\ x^2: \quad 2/3 &= 2w_1x_1^2 \\ x^4: \quad 2/5 &= 2w_1x_1^4 \end{aligned}$$

The ratio of the last two equations gives

$$\begin{aligned} x_1^2 &= 2/5 \div 2/3 = 3/5 \\ x_1 &= -\sqrt{3/5} \\ x_3 &= \sqrt{3/5} \end{aligned}$$

From the second equation

$$w_1 = \frac{1}{3x_1^2} = \frac{5}{9}$$

and from the first $w_2 = 8/9$. The use of symmetry clearly makes things easier!

PROBLEMS

19.2.1 Derive $\int_{-1}^1 f(x) dx = f(-1/\sqrt{3}) + f(1/\sqrt{3})$ $E_4 = 8/45$
(See Sec. 16.5.)

19.2.2 Derive $-\int_0^1 \ln x f(x) dx = w_1 f(x_1) + w_2 f(x_2)$

19.2.3 Derive $\int_0^1 x f(x) dx = w_1 f(x_1) + w_2 f(x_2)$ $x_i = \frac{5 \pm \sqrt{40}/7}{9}$

19.2.4 Derive $\int_{-\pi/2}^{\pi/2} \sin x f(x) dx = w_1 f(x_1) + w_2 f(x_2)$

19.2.5 Derive $\int_{-1}^1 f(x) dx = w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3) + w_4 f(x_4)$

Use symmetry to show that $w_1 = w_4$, $x_1 = -x_4$, $x_2 = -x_3$, and $w_2 = w_3$, so that $\pi(x)$ is a quadratic in x^2 .

19.2.6 Derive $\int_{-\infty}^{\infty} e^{-x^2} f(x) dx = w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)$

Use symmetry to show that $w_1 = w_{-1}$, $x_1 = -x_3$, and $x_2 = 0$.

19.2.7 Derive $\int_{-\pi}^{\pi} \sin x f(x) dx = \sum_{i=1}^4 w_i f(x_i)$

Use odd symmetry to show that $w_1 = -w_4$ and $w_2 = -w_3$.

$$19.2.8 \quad \text{Derive } \int_{-1}^1 xf(x) dx = \sqrt{5/27}[f(-\sqrt{3/5}) + f(\sqrt{3/5})]$$

$$19.2.9 \quad \text{Derive } \int_{-1}^1 \sqrt{1-x^2}f(x)dx = \frac{\pi}{4}[f(-1/2) + f(1/2)]$$

19.3 GAUSS' QUADRATURE (INTEGRATION)—FORMAL

The best-known case of using the sample positions as parameters in an integration formula is Gauss' quadrature on N points:

$$\int_{-1}^1 f(x) dx = \sum_{k=1}^N w_k f(x_k)$$

where both the weights and the sample points are regarded as parameters. Thus the examples in the previous section were Gaussian quadratures. With $2N$ parameters we can write down the $2N$ defining equations:

$$\begin{aligned} 1: \quad \int_{-1}^1 1 \cdot dx &= 2 = m_0 = w_1 + w_2 + \cdots + w_N \\ x: \quad \int_{-1}^1 x dx &= 0 = m_1 = w_1 x_1 + w_2 x_2 + \cdots + w_N x_N \\ x^2: \quad \int_{-1}^1 x^2 dx &= \frac{2}{3} = m_2 = w_1 x_1^2 + w_2 x_2^2 + \cdots + w_N x_N^2 \\ &\dots \dots \dots \\ x^{2N-1}: \quad \int_{-1}^1 x^{2N-1} dx &= 0 = m_{2N-1} = w_1 x_1^{2N-1} + w_2 x_2^{2N-1} + \cdots + w_N x_N^{2N-1} \end{aligned}$$

We again use the polynomial

$$\begin{aligned} \pi(x) &= (x - x_1)(x - x_2) \cdots (x - x_N) \\ &= x^N + c_{N-1}x^{N-1} + \cdots + c_0 \end{aligned}$$

We now multiply the j th equation by c_j and sum to get

$$\sum_{j=0}^{N-1} m_j c_j + m_N = \sum_{k=1}^N w_k \pi(x_k) = 0$$

If we shift the multipliers c_j down one line and repeat, we get

$$\sum_{j=0}^{N-1} m_{j+1} c_j + m_{N+1} = \sum_{k=1}^N w_k x_k \pi(x_k) = 0$$

Repeating this shifting process, we are finally led to a total of N equations

$$\sum_{j=0}^{N-1} m_{j+k} c_j + m_{N+k} = 0 \quad k = 0, 1, \dots, N-1$$

In the next section we will show that under very reasonable assumptions the persymmetric determinant [44] of these equations is not zero, and hence the system can be solved for the c_j . We then put these c_j into the equation for the sampling places

$$\pi(x) = x^N + c_{N-1}x^{N-1} + \cdots + c_0 = 0$$

This will, as we show in the next section, yield N real, distinct zeros. With these zeros we are now reduced to the case in Chap. 15 with known sample points, and we can solve the first N of the defining equations for the weights. Thus we have our formula.

An examination of the process shows that we first solve N simultaneous linear equations involving the moments m_k to obtain the c_j . These c_j are then used to form a polynomial, and we have to find the N zeros of it. Using these N zeros, we now solve another set of N simultaneous linear equations for the N weights. If all goes well, we will have our formula. We next turn to showing that all will go well in certain important cases.

19.4 GAUSS' QUADRATURE—ANALYSIS

The purpose of this section is to show that for a wide class of formulas the process of finding a Gaussian integration formula will go through without a hitch, in the sense that mathematically everything will be as we said it would be. Consider the class of integration formulas

$$\int_a^b K(x)f(x) dx = \sum_{k=1}^N w_k f(x_k)$$

where we assume that

$$K(x) \geq 0 \quad a \leq x \leq b$$

The moments are

$$m_j = \int_a^b K(x)x^j dx$$

The defining equations are:

$$m_j = \sum_{k=1}^N w_k x_k^j \quad j = 0, 1, \dots, 2N-1$$

If we multiply the j th equation by the c_j defined by

$$\begin{aligned} \pi(x) &= (x - x_1)(x - x_2) \cdots (x - x_N) \\ &= x^N + c_{N-1}x^{N-1} + \cdots + c_0 \quad c_N = 1 \end{aligned}$$

and sum, we get

$$\sum_{j=0}^{N-1} m_j c_j + m_N = \sum_{k=0}^N w_k \pi(x_k) = 0$$

Shifting and repeating, we get, in all, the system of equations as before

$$\sum_{j=0}^{N-1} m_{j+k} c_j + m_{N+k} = 0 \quad k = 0, 1, \dots, N-1$$

To prove that the determinant of these equations is not zero, we show that the only solution the *homogeneous* equations can have is $c_j = 0$ for all j . Using the set of N *homogeneous* equations, we multiply the k th equation by c_k and add, to get

$$\sum_{k=0}^{N-1} \sum_{j=0}^{N-1} m_{j+k} c_j c_k = 0$$

Recalling the definition of m_k , we have

$$\int_a^b K(x) \left[\sum_{k=0}^{N-1} c_k x^k \sum_{j=0}^{N-1} c_j x^j \right] dx = 0$$

or

$$\int_a^b K(x) \left[\sum_{k=0}^{N-1} c_k x^k \right]^2 dx = 0$$

Since $K(x) \geq 0$, the above equation can be zero only if $\sum c_k x^k \equiv 0$, that is, when all the $c_j = 0$ (as determined by the homogeneous equations). This in turn means that the determinant cannot be zero, and hence that we can solve the original set of nonhomogeneous linear equations for the c_j .

We now have the polynomial $\pi(x)$ whose zeros are the sample points. We need to show that when $K(x) \geq 0$, this particular polynomial has exactly N real, distinct zeros in the interval $a \leq x \leq b$. We assume, to the contrary, that it does not, that possibly there are complex zeros, real zeros outside the interval, or multiple zeros. Take all the odd-order zeros in the interval and for each write a simple factor in the product

$$p(x) = (x - x_1)(x - x_2) \cdots (x - x_k)$$

By our assumption (that, we will prove, leads to a contradiction) the degree k of $p(x)$ is less than N . Consider now

$$\int_a^b K(x) \pi(x) p(x) dx = \sum_{j=1}^N w_j \pi(x_j) p(x_j) = 0$$

The right-hand side of the formula is zero because $\pi(x_j) = 0$ at all the sample points. Furthermore we made the formula exact for all powers of x up to $2N$ and hence for all polynomials of degree less than $2N$. But $\pi(x)p(x)$ is of degree less

than $2N$, and so the integral must be zero. Finally, the integrand is always non-negative, and we have a contradiction since

$$\int_a^b K(x)\pi(x)p(x) dx > 0$$

To avoid the contradiction, we must have $k = N$, and thus there are N real, distinct zeros in the interval $a \leq x \leq b$.

When we solve the first N defining equations for the weights w_j (the Vandermonde determinant is not zero), we can reasonably wonder if they will also be appropriate for the last N equations. The equations that defined the c_i show how the higher moments are linearly related to the lower moments, and we can apply these relations to show that the last N equations are consistent with the first N since each (after the first N) is a linear combination of the earlier ones.

It is an important fact that when $K(x) \geq 0$, the weights w_i are all positive numbers. This follows from picking $f(x) = \pi_i^2(x)$ which is a polynomial of degree $2(N-1)$ for which the error is zero. We get

$$\int_a^b K(x)\pi_i^2(x) dx = \pi_i^2(x_i)w_i$$

and hence $w_i > 0$. Thus roundoff propagation is not severe in these formulas (see Sec. 19.8).

19.5 THE ERROR TERM

It would clearly be awkward to prove directly that the associated $G(s)$ is of constant sign, and so we proceed in an indirect fashion. Consider the Hermite interpolation formula (Sec. 17.2) using the N sample points. We have

$$f(x) = P_{2N-1}(x) + \pi^2(x) \frac{f^{(2N)}(\theta)}{(2N)!} \quad \theta = \theta(x)$$

We know that the integration formula is exact for all polynomials of degree less than $2N$, and so the polynomial part of the Hermite interpolation formula will exactly cancel out on both sides of the formula. That leaves the error term to be considered. On the right-hand side it cancels out due to the factor $\pi(x)$. On the left we have

$$\int_a^b K(x)\pi^2(x) \frac{f^{(2N)}(\theta)}{(2N)!} dx \quad K(x) \geq 0$$

where $\theta = \theta(x)$. Note that the factor $K(x)\pi^2(x)$ is of constant sign, and so we may apply the mean value theorem for integrals (Sec. 4.7) and get

$$\frac{f^{(2N)}(\theta^*)}{(2N)!} \int_a^b K(x)\pi^2(x) dx$$

But the integral is the same as

$$\int_a^b K(x)\pi^2(x) dx = E_{2N} \neq 0$$

since all the lower powers of x do not contribute to the error. Thus we have for the error of Gauss' quadrature, when $K(x) \geq 0$,

$$\frac{E_{2N}}{(2N)!} f^{(2N)}(\theta)$$

We earlier proved (Sec. 16.7) that if the form of the error was as indicated, then the $G(s)$ must be of constant sign.

All this reasoning has been based on the assumption that in the interval, $K(x)$ does not change sign. If it does change sign, then all kinds of pathological cases can arise. The extreme case is perhaps the kernel

$$K(x) = e^{-x^{1/4}} \sin x^{1/4}$$

For the integral

$$\int_0^\infty e^{-x^{1/4}} \sin x^{1/4} f(x) dx$$

all the moments

$$m_k = \int_0^\infty e^{-x^{1/4}} \sin(x^{1/4}) x^k dx = 0$$

by Sec. 13.7. This shows that the defining equations are satisfied if all the weights are zero, and it gives rise to the formula

$$\int_0^\infty e^{-x^{1/4}} \sin x^{1/4} f(x) dx = 0$$

which is true for all polynomials (of finite degree). But for the function

$$f(x) = \sin x^{1/4}$$

the integral clearly has a positive value. Fortunately in practice the kernel $K(x)$ of the integral is usually of constant sign, so that the above analysis is relevant and the particular example is not.

PROBLEMS

19.5.1 Discuss the kernel

$$K(x) = e^{-x^{1/4}}[1 + \sin x^{1/4}] \quad 0 \leq x \leq \infty$$

which does not change sign. See [27].

19.6 THREE SPECIAL CASES

There are three important special cases for which the sample points x_i and the corresponding weights w_i are known in a closed, analytical form. For these cases there is no necessity for reading in these numbers from some other source—they can be programmed in. These cases are:

$$\begin{aligned} \int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx &= \frac{\pi}{N} \sum_{k=1}^N f\left(\cos \frac{2k-1}{2N} \pi\right) + \frac{\pi}{2^{2N-1}} \frac{f^{(2N)}(\theta)}{(2N)!} \\ \int_{-1}^1 \sqrt{1-x^2} f(x) dx &= \frac{\pi}{N+1} \sum_{k=1}^N \sin^2 \frac{\pi k}{N+1} f\left(\cos \frac{k\pi}{N+1}\right) + \frac{\pi}{2^{2N}} \frac{f^{(2N)}(\theta)}{(2N)!} \\ \int_{-1}^1 \frac{\sqrt{1-x}}{\sqrt{1+x}} f(x) dx &= \frac{4\pi}{2N+1} \sum_{k=1}^N \sin^2 \frac{k\pi}{2N+1} f\left(\cos \frac{2k\pi}{2N+1}\right) + \frac{\pi}{2^{2N}} \frac{f^{(2N)}(\theta)}{(2N)!} \end{aligned}$$

We shall derive the first case [30]. The defining equations are:

$$\int_{-1}^1 \frac{x^m}{\sqrt{1-x^2}} dx = \sum_{j=1}^N w_j x_j^m \quad m = 0, 1, \dots, 2N-1$$

Setting

$$x = \cos \theta \quad x_i = \cos \theta_i$$

then

$$\int_0^\pi \cos^m \theta d\theta = \sum_{j=1}^N w_j \cos^m \theta_j$$

We know

$$\begin{aligned} \cos N\theta &= \text{polynomial of degree } N \text{ in } \cos \theta \\ &= \text{a linear combination of } (\cos \theta)^m \end{aligned}$$

Taking exactly that same linear combination of the first $N+1$ defining equations, we get

$$\int_0^\pi \cos N\theta d\theta = \sum_{i=1}^N w_i \cos N\theta_i$$

With small modifications of this trick we can get the $2N$ equations

$$\begin{aligned}\int_0^\pi \cos^m \theta \, d\theta &= \sum_{j=1}^N w_j \cos^m \theta_j \\ \int_0^\pi \cos N\theta \cos^m \theta \, d\theta &= \sum_{j=1}^N w_j \cos N\theta_j \cos^m \theta_j\end{aligned}\quad m = 0, 1, \dots, N-1$$

which are equivalent to the original $2N$ equations.

Since $\cos m\theta$ equals the polynomial in $\cos \theta$, we can similarly convert these equations to the equivalent set (see Sec. 26.2):

$$\begin{aligned}\int_0^\pi \cos m\theta \, d\theta &= 0 = \sum_{j=1}^N w_j \cos m\theta_j \\ \int_0^\pi \cos N\theta \cos m\theta \, d\theta &= 0 = \sum_{j=1}^N w_j \cos N\theta_j \cos m\theta_j\end{aligned}$$

The second N equations will be satisfied if

$$\cos N\theta_j = 0$$

or

$$\theta_j = \frac{2j-1}{2N} \pi \quad j = 1, 2, \dots, N$$

From the orthogonality of the cosines over the equally spaced θ_j (Prob. 11.1.3), we see that the first N equations will be satisfied if $w_j \equiv w$ and from the first equation that

$$w = \frac{\pi}{N}$$

Thus we have our formula.

Similar, more elaborate trigonometry will derive the other two formulas, although there are other, perhaps preferable, ways to find them.

PROBLEMS

19.6.1 Derive the second formula.

19.6.2 Derive the third formula.

19.7 GIVEN SOME SAMPLE POINTS

It often happens that the values of the function are known at one or both ends of the interval of integration. Formulas which take advantage of this are known as Radau and Lobatto integrations, respectively.

The effect of fixing a sample point is to decrease the number of defining equations by 1. Let us examine the case of one endpoint, say at $x = a$. For the formula

$$\int_a^b K(x)f(x) dx = w_a f(a) + \sum_{j=1}^{N-1} w_j f(x_j)$$

we have the defining equations

$$m_k = w_a a^k + \sum_{j=1}^{N-1} w_j x_j^k \quad k = 0, 1, \dots, 2N-2$$

If we multiply the k th equation ($k = 0, 1, \dots, 2N-3$) by a and subtract from the $(k+1)$ st equation, we get

$$m_{k+1} - am_k = \sum_{j=1}^{N-1} w_j (a - x_j) x_j^k \quad k = 1, 0, \dots, 2N-3$$

Set

$$\begin{aligned} m_{k+1} - am_k &= M_k \\ w_j(a - x_j) &= W_j \end{aligned}$$

and we have, in effect, eliminated the known point and are reduced to the pure Gaussian case of all unknown points. Evidently we can iterate the process and eliminate another given sample point each time until all we have left are the unknown sample points. We then find the unknown x_i . With these we are then back to the earlier case of all sample points being known.

Again we may ask, How does it go in practice? Will the sample points be real, distinct, and in the interval? For the two cases, a sample point at one end of the interval (Radau) or a sample point at each end (Lobatto) and for $K(x) \geq 0$ ($a \leq x \leq b$), we can carry out all the steps. Corresponding to the analysis in Sec. 19.4 where we used $\pi^2(x)$, we now use

$$(x - a)\pi^2(x)$$

[where $\pi(x)$ is the product over the unknown sample points] in the case of one endpoint, and

$$(x - a)\pi^2(x)(b - x)$$

in the case of two endpoints. Thus if we can find the c_i , then the zeros of the polynomial are real, distinct, and in the interval $a \leq x \leq b$. The analysis that we can solve for the c_i is a bit messy, but follows readily from Sec. 19.4.

PROBLEMS

Find (where only endpoints are given):

$$19.7.1 \quad \int_{-1}^1 f(x) dx = \frac{1}{3}[f(-1) + f(1) + 5f(-\sqrt{1/5}) + 5f(\sqrt{1/5})] \quad E_6 = -2/3^3 \cdot 5^3 \cdot 7$$

$$19.7.2 \quad \int_0^\infty e^{-x} f(x) dx = \frac{1}{2}[f(0) + f(2)]$$

$$19.7.3 \quad \int_0^\infty e^{-x} f(x) dx = w_0 f(0) + w_1 f(x_1) + w_2 f(x_2)$$

$$19.7.4 \quad -\int_0^1 f(x) \ln x dx = w_0 f(0) + w_1 f(1) + w_2 f(x_2)$$

$$19.7.5 \quad \int_0^\infty e^{-x} f(x) dx = w_0 f(0) + w_1 f(x_1) + w_2 f(4) = \frac{1}{12}[3f(0) + 8f(1) + f(4)]$$

19.7.6 Carry out the details of the analysis in the case of Lobatto integration.

19.8 CHEBYSHEV INTEGRATION

It sometimes happens that we are either given or would like to put linear restraints on the weights of a formula. Such restraints are likely to involve the weights associated with the known sample points, since they are not easily applied to positions that have not even been specified in advance. The main exception is the important case of the same restraint applied to all the weights of the sample. The reason that this restraint arises is that this is the one that minimizes the effect of the noise in the samples; all other sets have some increase in the variance of the weights since the sum of the weights is given by the first moment m_0 . The proof goes as follows. To minimize

$$\sum_{i=1}^N w_i^2$$

subject to the restraint (the first defining equation) that

$$\sum_{i=1}^N w_i = m_0$$

we use Lagrange multipliers and minimize the function of the weights

$$F(w_i) \equiv \sum_{i=1}^N w_i^2 - \lambda \sum_{i=1}^N w_i$$

We get

$$\frac{\partial F}{\partial w_j} = 2w_j - \lambda = 0$$

$$w_i = \frac{2}{\lambda} = \frac{m_0}{N} \quad \text{for all } i$$

When all the weights are equal, the formula is called "Chebyshev."

The best-known Chebyshev case is $K(x) = 1$ with the range $-1 \leq x \leq 1$. The $w_i = w = 2/N$. The defining equations are:

$$m_k = w \sum_{i=1}^N x_i^k \quad k = 0, 1, \dots, N$$

or

$$\sum_{i=1}^N x_i^k = \frac{m_k}{w}$$

The sums

$$s_k = \sum_{i=1}^N x_i^k = \frac{m_k}{w}$$

are symmetric functions in the x_i and hence by the fundamental theorem [60] of symmetric functions theory are expressible in terms of the elementary symmetric functions

$$\begin{aligned} p_1 &= -\sum x_i \\ p_2 &= \sum x_i x_j \\ &\dots\dots\dots \\ p_N &= (-1)^N x_1 x_2 \cdots x_N \end{aligned}$$

Indeed, the Newton identities

$$\begin{aligned} s_1 + p_1 &= 0 \\ s_2 + p_1 s_1 + 2p_2 &= 0 \\ s_3 + p_1 s_2 + p_2 s_1 + 3p_3 &= 0 \\ &\dots\dots\dots \\ s_N + p_1 s_{N-1} + \cdots + Np_N &= 0 \end{aligned}$$

express exactly this relationship.

Thus, since the p_i are our c_{N-i} , we can convert these defining equations to equations in terms of the c_i . It is known that these equations can be solved for real x_i only for the cases¹ of $N = 1, 2, \dots, 7, 9$; all other cases give complex x_i .

But let us be reasonable about this; the complex zeros will occur, in this and other cases, only as conjugate pairs (we are considering integrating only real functions), and the weights will be such that the sum is real. Therefore, for the labor of finding one complex function value we get the accuracy equivalent, in some sense, to that of two real points. And why, given a mathematically expressed

¹ For tables to 10 decimal places see National Bureau of Standards, "Handbook of Mathematical Functions," *op. cit.*

integrand, should we refuse to use complex sample points? An experiment integrating a function with a known analytic answer for the cases $N = 1, 2, \dots, 12$ showed that the errors fell on a smooth curve, as expected.

The Chebyshev case

$$K(x) = \frac{1}{\sqrt{1-x^2}}$$

is given in Sec. 19.6 and can be solved for all N with real samples—indeed it is both a Gaussian and a Chebyshev formula at the same time!

19.9 RALSTON INTEGRATION

A. Ralston has published¹ an interesting example of a mixed type of integration in which two sample points and one linear restraint are assigned. Specifically, the formula

$$\int_{-1}^1 f(x) dx = \sum_{i=1}^n w_i f(x_i)$$

has the conditions

$$x_1 = -1$$

$$x_n = 1$$

$$w_1 = -w_n$$

We have, therefore, $2N - 3$ parameters and can make the formula exact for $1, x, \dots, x^{2N-4}$. The defining equations are

$$\begin{aligned} 2 &= \sum_{i=2}^{n-1} w_i \\ 0 &= -2w_1 + \sum w_i x_i \\ 2/3 &= \sum w_i x_i^2 \\ 0 &= -2w_1 + \sum w_i x_i^3 \\ &\dots\dots\dots \\ \frac{2}{2N-3} &= \sum w_i x_i^{2N-4} \end{aligned}$$

In order to follow the subsequent algebra more easily and at the same time not oversimplify it, we take the case of $N = 5$ which gives $2N - 3 = 7$ equations.

¹ A. Ralston, *J. Assoc. Computing Machinery*, vol. 6, pp. 384–394, July, 1959.

The sample polynomial is

$$\pi(x) = (x - x_2)(x - x_3)(x - x_4) = x^3 + c_2 x^2 + c_1 x + c_0$$

The usual elimination process leads to $N - 1$ (in this case, four) equations:

$$\begin{array}{rclcl} c_0 & + \frac{1}{3}c_2 & = & -w_1 c_1 & -w_1 \\ \frac{1}{3}c_1 + \frac{1}{5} & = & -w_1 c_0 & -w_1 c_2 & \\ \frac{1}{3}c_0 & + \frac{1}{5}c_2 & = & -w_1 c_1 & -w_1 \\ \frac{1}{5}c_1 + \frac{1}{7} & = & -w_1 c_0 & -w_1 c_2 & \end{array}$$

It is easy to see that we can form $N - 3 (= 2)$ equations which do not involve the quadratic terms $w_1 c_i (i = 0, 1, 2)$,

$$\begin{aligned} \frac{2}{3}c_0 + \frac{2}{15}c_2 &= 0 \\ \frac{2}{15}c_1 + \frac{2}{35} &= 0 \end{aligned}$$

We have, therefore, $N - 3$ linear, nonhomogeneous equations in the unknowns $c_0, c_1, \dots, c_{N-3}$, or one more unknown than equations. Solving the equations in terms of, say, c_0 , we have

$$\begin{aligned} c_1 &= -3/7 \\ c_2 &= -5c_0 \end{aligned}$$

These are put in the first two quadratic equations of the set to give

$$\begin{aligned} -\frac{1}{3}c_0 &= -\frac{2}{7}w_1 \\ \frac{1}{35} &= 2c_0 w_1 \end{aligned}$$

Thus in the general case, as well as in the specific case of $N = 5$, when we eliminate c_0 , we come to a quadratic in w_1 . In the case of $N = 5$,

$$w_1 = \pm \sqrt{1/60}$$

The plus or minus sign has the effect of reversing the formula from left to right or right to left.

Knowing w_1 , we can easily find the c_i and the sample polynomial

$$\pi(x) = x^3 - 5\left(\frac{3}{245}\right)x^2 - \frac{3}{7}x + \frac{3}{245}$$

Reasoning similar to that of Sec. 19.4 shows that the zeros are real, distinct, and in the interval of integration.

PROBLEMS

19.9.1 Show that the Ralston case of $N = 6$ has the sample polynomial

$$\pi(x) = x^4 + \frac{4x^3}{3\sqrt{6}} - \frac{2}{3}x^2 - \frac{4x}{7\sqrt{6}} + \frac{1}{21}$$

19.10 GAUSSIAN INTEGRATION USING DERIVATIVES

It is natural to turn to Gaussian integration using both function and derivative values. If only the function and first-derivative values are used, then the sample points often (not always) turn out to be complex; but if the function and the first and second derivatives are used, then much of the analytical part of Sec. 19.4 can be carried through using $\pi^4(x)$ in place of $\pi^2(x)$.

Rather than examine the messy general case, let us examine the particular case

$$\begin{aligned} \int_{-1}^1 f(x) dx = & w_1 f(-x_1) + w_0 f(0) + w_1 f(x_1) \\ & - w'_1 f'(-x_1) + w'_1 f'(x_1) \\ & + w''_1 f''(-x_1) + w''_0 f''(0) + w''_1 f''(1) \end{aligned}$$

where we have used symmetry to reduce the complexity of the algebra and have made it exact for *all* odd powers of x .

The defining equations are, on dividing by 2,

1:	1	=	w_1	+ $1/2 w_0$		0
x^2 :	$1/3$	=	$w_1 x_1^2$	+ $2w'_1 x_1 + 2w''_1$	+ w''_0	0
x^4 :	$1/5$	=	$w_1 x_1^4$	+ $4w'_1 x_1^3 + 12w''_1 x_1^2$		+ x_1^6
x^6 :	$1/7$	=	$w_1 x_1^6$	+ $6w'_1 x_1^5 + 30w''_1 x_1^4$		- $3x_1^4$
x^8 :	$1/9$	=	$w_1 x_1^8$	+ $8w'_1 x_1^7 + 56w''_1 x_1^6$		+ $3x_1^2$
x^{10} :	$1/11$	=	$w_1 x_1^{10}$	+ $10w'_1 x_1^9 + 90w''_1 x_1^8$		- 1

Multiplying these equations by the constants on the right and adding, we get (dropping the subscripts on the x)

$$\frac{x^6}{5} - \frac{3}{7}x^4 + \frac{3}{9}x^2 - \frac{1}{11} = 0$$

Regarding this as a function of x ,

$$f(x) \equiv \frac{x^6}{5} - \frac{3}{7}x^4 + \frac{3}{9}x^2 - \frac{1}{11} = \text{a cubic in } x^2$$

we see that $f(0) = -1/11$ and $f(1) = 8(1/33 - 1/35) > 0$, and hence there is a real x that is the solution to our problem of finding a sample point. The rest now follows easily.

PROBLEMS

- 19.10.1 Carry out the details of the analysis mentioned in the text.
 19.10.2 Complete the above special case.
 19.10.3 Discuss the case where some points use $f(x)$, some $f(x), f'(x), f''(x)$, and some $f, f', f'', f''', f^{(4)}$.
 19.10.4 Show that

$$\int_{-1}^1 f(x) dx = w_1 f(x_1) + w_2 f(x_2) + w'_1 f'(x_1) + w'_2 f'(x_2)$$

leads to complex sample points x_i .

- 19.10.5 Derive the two-point case using f, f' , and f'' for $\int_{-1}^1 f(x) dx$.

19.11 AN ALGORITHMIC APPROACH TO FINDING FORMULAS

Computer science emphasizes the algorithmic approach to various fields, and we shall adopt it for the problem of finding formulas. In order not to obscure matters too much we shall omit a lot of the trivial details that are easily supplied (especially exits from the flow diagram when various steps fail to work).

The formulas we are considering are of the form of a linear operator which is being estimated by a linear combination of samples of the function, and possibly one or more derivatives, using weights $w_i, w'_i, w''_i, \dots$. We have examined the problem mainly in terms of the operation of integration over a definite interval, but much of the material applies to *any* linear operator.

To fix notation, let

- n_1 = the number of weights $w_i, w'_i, w''_i, \dots$ to be determined
- n_2 = the number of sample points x_i that are given
- n_3 = the number of sample points x_i to be found
- n_4 = the number of linear restraints on the weights
- $N = n_2 + n_3$

We are finding formulas in the classical sense of making them correct for successive powers of x , that is, putting in $1, x, x^2, \dots$ as far as we can to form the defining equations, Sec. 15.5. How far can we go? We have $n_1 + n_3 - n_4$ free parameters and can therefore expect that the same number of defining equations will *usually* determine all the parameters. We will, therefore, need the same number of moments m_i of the linear operator, and later we will need more m_i , both for the possibility of higher accuracy (as, for example, in Simpson's formula, Sec. 15.3) and for the error term, Chap. 16.

Before writing down the defining equation, box 1 in Fig. 19.11.1, it is well to look for symmetry about some point $x = a$, either even or odd symmetry (Sec. 17.5, Example 17.3, and Sec. 19.2). If there is symmetry, then using powers of $x - a$, instead of powers of x , will simplify the problem (though it should be realized that anything that can be done along these lines can be done later by manipulating the equations properly). When we use the symmetry we have:

Even symmetry	Odd symmetry
$m_{2k+1} = 0 \quad \text{all } k$	$m_{2k} = 0 \quad \text{all } k$
$x_i - a = -(x_{N-i} - a)$	$x_i - a = -(x_{N-i} - a)$
$w_i = w_{N-i}$	$w_i = -w_{N-i}$
$w'_i = -w'_{N-i}$	$w'_i = w'_{N-i}$
$w''_i = w''_{N-i}$	$w''_i = -w''_{N-i}$
.....	

These choices make the corresponding defining equations exact for *all* values of the index k , so that we need only write the defining equations for

$$x^{2k} \quad \text{or} \quad x^{2k+1}$$

Thus there will be about half as many defining equations to be solved and about one-eighth as much algebraic manipulation to solve them.

To the defining equations we append the n_4 equations giving the linear restraints on the weights. The combined set of $n_1 + n_3$ equations are to be solved for both the x_i that are not known and for the weights $w_i, w'_i, w'', \dots$.

If $n_3 = 0$ (all sample points are known), then we solve these equations (box 2 in Fig. 19.11.1). There are three approaches possible:

- 1 Direct elimination—the hard way.
- 2 Solve for some of the weights using the sampling polynomials π_i, σ_i, τ_i , etc., to obtain them directly, Secs. 15.5, 17.4, and 17.6.
- 3 In the Birkhoff cases we can construct the inverse matrix as combinations of sums and products of the x_i (Secs. 14.5, 14.6, 17.4, 17.6).

These equations can be solved when the system has Birkhoff data (consecutive derivatives) (Sec. 17.6). In other cases we are not certain that they can be solved; indeed we know from the example of Sec. 17.7 that they can be inconsistent or that they can admit of a one-parameter (or more) family of solutions. In the later case we can add further defining equations and hope for consistency when the rank rises to full order (Sec. 17.7).

When we have the formula, we next turn to finding the error term (box 3 of Fig. 19.11.1) which requires additional moments m_i . We first find the value of m such that $E_m \neq 0$. Note that the value of E_m can be found *directly*, without finding the weights, as a linear combination of the moments m_i . The coefficients of this linear combination are polynomials in the x_i only (Sec. 17.8). We then construct the $G(s)$, and if it is of constant sign, we have the structure of the error

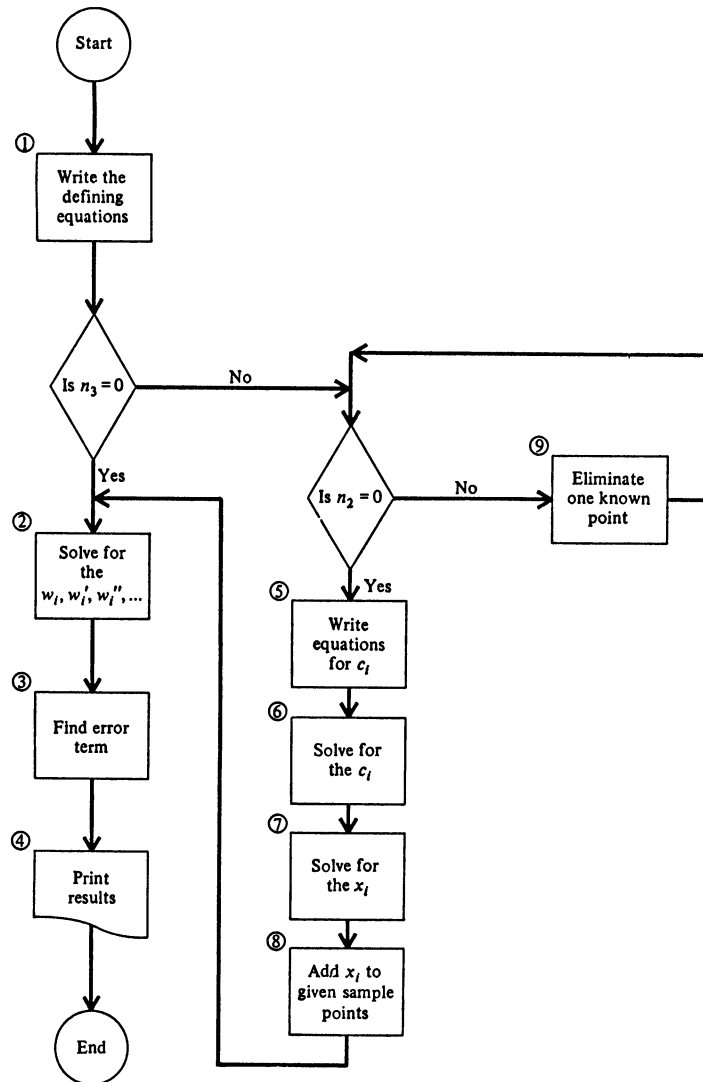


FIGURE 19.11.1

term (Sec. 16.5), while if it is not of constant sign, we turn to Sec. 16.7 for guidance. For some Gaussian cases we can use the interpolation formula's error term instead (Sec. 19.5).

Finally it is necessary to organize our scattered results and print out the formula, or why we did not get it (box 4 in Fig. 19.11.1).

If $n_3 \neq 0$ but $n_2 = 0$, we have the pure Gaussian case of all unknown sample points. For the case of only function values we first construct the equations for the coefficients c_i of the polynomial $\pi(x)$ whose zeros are the unknown sample points. This we can always do (box 5 in Fig. 19.11.1).

To solve these equations for the c_i is another matter (box 6 in Fig. 19.11.1). We need to consider the rank of the equations and what to do if this is less than full rank. Adding more defining equations is one approach, provided inconsistency has not shown that such a formula cannot exist. For the case of $K(x) \geq 0$, integration with no constraints on the weights, and at most only endpoints as assigned samples, we know that the equations are of full rank (Sec. 19.4). The general case using derivatives is obscure.

In box 7 of Fig. 19.11.1, we solve this polynomial for the unknown sample points x_i . Again we are assured that in the case of integration with $K(x) \geq 0$, and at most assigned sample points at the ends of the interval, we will find real, distinct zeros in the interval of integration. This can happen in other cases too, and it can fail to happen. But the conventional view that complex zeros or multiple zeros are always fatal should be resisted, and the question judged on its own merits. It is easy to show that in the case of complex zeros only the corresponding weights can be complex. The complex sample points give rise to a real quadratic factor in the $\pi(x)$ and most of the $\pi_i(x)$; and almost all the usual proofs go through as before, with, at most, minor modification. If multiple roots occur, then thought should be given to using a one-higher derivative at that place.

In box 8 of Fig. 19.11.1, we add these new sample points to the old ones (if any) and look for duplicates. If there are none, then using a suitable number of defining equations, we carry out box 2 (Fig. 19.11.1) and know that the other defining equations will automatically be satisfied. Duplicate points are unlikely and have not been discussed.

If $n_3 \neq 0$, then $n_2 \neq 0$, and we go to box 9 (Fig. 19.11.1) and eliminate a given sample point (Sec. 19.7). Going around the loop enough times, we end up with all unknown sample points x_i and go to box 5.

We have ignored many details, and much research can be done to increase the number of circumstances under which we can be sure that the method will go through without any problems in going from the defining equations to the formula. With modern symbol-manipulating routines available, this process could be put on a machine.

20.1 INTRODUCTION

The previous Chaps. 14 to 19 examined the process of approximating a linear operator as a linear combination of samples of the function, and possibly some its derivatives, and used integration as a typical linear operator. These chapters concentrated on *methods*, as the title of the book suggests, and did not attempt to apply the resulting formulas to specific functions. The present chapter looks a little closer at the actual application of the methods to the practical problem of integration, although we shall also look once more at the problem of interpolation. In both cases we will look at how formulas can be combined (composite formulas) to achieve the desired result.

Books on numerical methods often give many examples of formulas applied to specific functions. We do not intend to do this for a number of reasons. First, it supposes that the reader cannot evaluate a function, and possibly some of its derivatives, at a given set of sample points and then compute a weighted sum of these results—presumably drill in such matters is not necessary at this late stage in the reader's mathematical development. Second, for a given formula for integration it is easy to find a function for which it will give a very good result, while at the

same time a “better” formula will give a worse result. Formulas are *not* to be judged on how well they do on a specific function, but rather how well they do over some class of functions. We cannot use the *average* error over any reasonable ensemble of functions as a measure because, among other reasons, the ensemble is very likely to have both $f(x)$ and $-f(x)$ in it with equal probability, so that the average error will automatically be zero—we probably want to use the variance over the ensemble if only we could agree on one or more suitable ensembles of functions! In the absence of such tests, we are forced to use the size of the coefficient E_m in the error term as a guide for comparing formulas having the same-order error term (the same m). As noted earlier, we are not certain about the effect of the mean value theorem in placing the θ value more favorably in one formula than in another, and so our choice is not certain, only plausible. Third, with large-scale computation it is more and more true that the user does not get to look at the results—they are buried deep in some inner loop and at best he sees the results of a few early test cases. But worse than this, he is often forced to pick the method *before* any real data are available and later can modify his choice only slightly. Commitments in large projects must be made in advance of definite knowledge, and at much later times cannot be broken easily without jeopardizing the whole project. Thus it is necessary to train numerical analysts to know in advance what will probably happen. Of course tests can and should often be made of specific cases. Of course somewhat similar problems can be tried before making the final choice, but many times there is no real hope of knowing more than a general description of what the data will be like when they become available.

For all these reasons we will stick to the somewhat abstract approach of trying to reason what will happen in practice rather than regularly appealing to specific examples—this approach is necessary in order to get training in this difficult art of being right even when you are not sure what the problem is. This puts a burden on testing whatever seems, at the moment, to be relevant material whenever you can. Unfortunately the design of experiments lies outside the course, and the art of guessing is very difficult to teach.

20.2 POLYNOMIAL APPROXIMATION AGAIN

By now it should be very clear that what we are doing by our formulas is equivalent (with a very few exceptions) to first finding an interpolating polynomial and then applying a linear operator to this interpolating polynomial—we called it analytic substitution. We need, therefore, to think a bit about what kinds of functions

can be approximated well by low-order polynomials, as well as the chief characteristics of high-order polynomials.

First, a polynomial is finite, along with all orders of its derivatives, at all finite points. This means that integrals like

$$\int_0^h \sqrt{x} e^{-x} dx \quad \int_0^h \frac{\arctan x}{\sqrt{x}} dx \quad - \int_0^h \ln x \arctan x dx$$

usually are not going to be very accurately done, except by chance or by h being small. One frequently sees proof in the literature that if a sufficiently small spacing is used, then some particular composite formula will give accurate results (in the limit), but it seems foolish to try to imitate the fundamental definition of integration by adding enough small intervals to get the sum accurately.

Second, polynomials go to infinity as the argument goes to both $+\infty$ and $-\infty$. Thus integrals of the form

$$\int_0^\infty \frac{dx}{1+x^2} \quad \int_e^\infty \frac{dx}{x^2 \ln x} \quad \int_0^\infty e^{-x} f(x) dx$$

are apt to be troublesome, unless the kernel smothers the growth at infinity, as is likely to be true in the third example.

Third, polynomials love to wiggle, especially high-order ones. But for integration with a constant kernel $K(x) = 1$, the wiggles are apt to be more awful-looking than they are in practice. There is a tendency for the successive loops to cancel out more than might seem reasonable at first sight. For example, in Simpson's formula applied to a cubic the area between the interpolating quadratic and the cubic had a net area of zero (Sec. 15.3).

The problem, then, is to discover the nature of any nonpolynomial behavior and to incorporate this into the kernel $K(x)$; it is foolish to do otherwise. A very high percentage of the problems that arise in practice will have some peculiar behavior at one or both ends of the interval, but only occasionally will there be trouble inside the range (being bounded at infinity is real trouble for a polynomial). It is because such troubles arise so often that we have given all the material for developing formulas so that an appropriate one can be found for almost any kind of difficulty you face. Only the most common cases, or perhaps the easiest ones to discuss, have been reported in the literature, and the others are awaiting you when you start practicing the difficult art of practical computing. Thus you will occasionally have to develop your own formulas to fit the peculiar needs of your current work.

20.3 THE NEWTON-COTES FORMULAS

The most obvious family of formulas, and the most commonly discussed, but not used in practice, is the Newton-Cotes formulas for integrating a function, given a set of equally spaced sample points (including the endpoints). The trapezoid rule (Sec. 16.5)

$$\int_0^h f(x) dx = \frac{h}{2} [f(0) + f(h)] - \frac{1}{12} h^3 f''(\theta)$$

Simpson's formula (Sec. 16.5),

$$\int_0^{2h} f(x) dx = \frac{h}{3} [f(0) + 4f(h) + f(2h)] - \frac{1}{90} h^5 f^{(4)}(\theta)$$

and the three-eighths rule (Prob. 15.4.4 in disguise)

$$\int_0^{3h} f(x) dx = \frac{3h}{8} [f(0) + 3f(h) + 3f(2h) + f(3h)] - \frac{3}{80} h^5 f^{(4)}(\theta)$$

are the first three members of the family. Table 20.3 gives the coefficients for the general form

$$\int_0^{nh} f(x) dx = Ah[B_0 f(0) + B_1 f(h) + B_2 f(2h) + \cdots + B_n f(nh)] + R_n$$

where n is the number of intervals, *not* the number of sample points as we have been doing. Evidently there are $n + 1$ sample points. Since the coefficients are symmetric, we need not give all the table.

Some of the coefficients become negative for $n = 8$. For $n = 9$ they are all positive, but for $n \geq 10$ there are negative coefficients. This tends to produce poor roundoff properties, and as a result the higher-order Newton-Cotes formulas are seldom used. The order of the error terms jumps by 2 in going from an odd number to the next even number, which tends to favor the even-order formulas.

Notice the appearance of the powers of h in the error term. When we use other than unit spacing, the powers of h automatically appear in the defining equations. We may either compute with the h values present or else pretend we have unit spacing and supply the proper power of h later (for equally spaced samples); the latter is usually preferable.

If these formulas are to be compared for the same interval, then it must be remembered that $h = \text{range of integration}/n$, and the corresponding adjustments must be made in the error terms

The Newton-Cotes formulas can be derived in many ways. Perhaps the simplest way to find the actual coefficients is to observe that the Gregory formula, when written to include all the differences that can be computed from the sample

Table 20.3

n	$\mathcal{A}$	B_0	B_1	B_2	B_3	B_4	B_5	R_n
1	$\frac{1}{2}$	1	1					$-(1/12)h^3 f'''(\theta)$
2	$\frac{1}{3}$	1	4	1				$-(1/90)h^5 f^{(5)}(\theta)$
3	$\frac{1}{6}$	1	3	3	1			$-(3/80)h^5 f^{(5)}(\theta)$
4	$\frac{1}{24}$	7	32	12	32	7		$-(8/945)h^7 f^{(7)}(\theta)$
5	$\frac{1}{120}$	19	75	50	50	75	19	$-(275/12,096)h^7 f^{(7)}(\theta)$
6	$\frac{1}{720}$	41	216	27	272	27	216	$-(9/1,400)h^9 f^{(9)}(\theta)$
7	$7/17,280$	751	3,577	1,323	2,989	2,989	1,323	$-(8,183/518,400)h^9 f^{(9)}(\theta)$
8	$4/14,175$	989	5,888	-928	10,496	-4,540	10,496	
9	$9/89,600$	2,857	15,741	1,080	19,344	5,778	5,778	
10	$5/299,376$	16,067	106,300	-48,525	272,400	-260,550	427,368	

points, is exact for polynomials of the maximum degree, and hence when written in the Lagrange form, must be the same as if it were derived directly. Thus for the case of $n = 4$ we have

$$\begin{aligned} \int_0^4 f(x) dx &= \frac{1}{2}f_0 + f_1 + f_2 + f_3 + \frac{1}{2}f_4 + \frac{1}{12}(\Delta f_0 - \Delta f_3) \\ &\quad + \frac{1}{24}(\Delta^2 f_0 + \Delta^2 f_2) + \frac{1}{720}(\Delta^3 f_0 - \Delta^3 f_1) \\ &\quad - \frac{3}{160}(\Delta^4 f_0 + \Delta^4 f_2) = \frac{1}{45}f_0 + \frac{6}{45}f_1 + \frac{2}{45}f_2 + \frac{6}{45}f_3 + \frac{1}{45}f_4 \end{aligned}$$

We have not derived the error terms [by showing that the influence function $G(s)$ is of constant sign] because as we said earlier, the formulas beyond Simpson's are seldom used.

PROBLEMS

- 20.3.1 Apply the $n = 2, 4, 6, 8$ Newton-Cotes cases to $\int_0^1 e^{-x} dx$. Compare with the correct answer.
- 20.3.2 Discuss the noise amplification of the Newton-Cotes formulas.
- 20.3.3 Derive the Newton-Cotes formula from the Gregory formula for $n = 6$.

20.4 REMARKS ON SOME FORMULAS

Why are the Newton-Cotes formulas seldom used for high orders? Partly because the roundoff appears to be so bad. If we assume that the roundoff error ε_i is independent at successive points x_i where the function is evaluated and that it has a noise level of σ^2 , then the variance of the sum will be

$$\begin{aligned} \text{Av}\{\sum w_i \varepsilon_i^2\} &= \sum_i \sum_j w_i w_j \text{Av}\{\varepsilon_i \varepsilon_j\} \\ &= (\sum w_i^2) \sigma^2 \end{aligned}$$

since

$$\text{Av}\{\varepsilon_i \varepsilon_j\} = 0 \quad i \neq j$$

But this effect should not be exaggerated, though it is serious when the weights have different signs since their sum is fixed in advance at m_0 .

The Gaussian formulas require peculiar weights and sample points, and this needlessly discourages people from using them; however the weights being all positive for $K(x) \geq 0$ is encouraging. The Chebyshev formulas do not seem to be worth the trouble of the awkward sample points *except* when they are given by

analytical formulas. Thus we are forced to look more closely at the composite formulas that were briefly mentioned in Sec. 18.9.

However, the Gaussian formulas are often very useful. Among other places they are often used in solving integral equations. For example, given the integral equation

$$y(x) = f(x) + \int_{-1}^1 K(x,s)y(s) ds$$

we apply Gaussian quadrature and are led to a system of simultaneous linear equations:

$$y(x_i) = f(x_i) + \sum_{j=1}^N w_j K(x_i, s_j) y(s_j) \quad i = 1, 2, \dots, N$$

Since the cost of solving linear equations is proportional to N^3 , the use of Gaussian quadrature, which reduces the number of equations by a factor of 2 for the same order of truncation error, is clearly worth the awkwardness of the peculiar sample points and weights.

20.5 COMPOSITE FORMULAS

Since high-order formulas are not often used, composite formulas must be the major formulas in practice. In the case of the Hamming-Pinkham formulas the error term is given by the sum of *all* the neglected differences and in practice is usually not far from the first neglected difference.

In the case of composite formulas based directly on some formula, say Simpson's, we find that the error term is the sum of derivatives, and we need the following theorem.

Theorem 20.5.1 If $g(x)$ is continuous and the $c_i \geq 0$, then for some value θ in the interval of all the arguments

$$g(\theta) \sum c_i = \sum_{i=1}^N c_i g(\theta_i)$$

PROOF The proof follows by induction from the simple case of two. Consider

$$\phi(\theta) \equiv c_1 g(\theta_1) + c_2 g(\theta_2) - (c_1 + c_2)g(\theta)$$

In the interval of the θ_i , $g(\theta)$ takes on both a maximum and a minimum. If $g(\theta)$ is not constant, then $\phi(\theta)$ has opposite signs at these two points: hence since it is continuous, it takes on the value 0. The induction follows easily. ////

This theorem is the discrete analog of the theorem

$$\int_a^b f(x)g(x) dx = g(\theta) \int_a^b f(x) dx \quad f(x) \geq 0$$

(Secs. 7.4 and 16.2). Thus we get for the composite Simpson's formula

$$\int_0^{2nh} f(x) dx = \frac{h}{3} [f(0) + 4f(h) + 2f(2h) + \cdots + f(2nh)] - \frac{nh^5 f^{(4)}(\theta)}{90}$$

The h^5 appears in the error term because we have shifted from unit spacing in t to the spacing h in x by the transformation

$$ht = x$$

$$dt = \frac{1}{h} dx$$

and

$$\frac{d^n}{dt^n} \equiv \frac{h^n d^n}{dx^n}$$

The error is often written as

$$- \frac{(b-a)h^4 f^{(4)}(\theta)}{45}$$

where $b-a = \text{range of integration} = 2nh$.

In formulas like Ralston's, the opposite signs of the function values at the ends clearly cause them to cancel out, except at the ends of the range, and this is their chief merit.

In formulas using derivatives there is a strong tendency to have the odd-order derivatives cancel out. Problem 17.5.2 derives the formula

$$\int_0^1 f(x) dx = \frac{1}{12} [6f(0) + 6f(1) + f'(0) - f'(1)] + \frac{f^{(4)}(\theta)}{720}$$

or

$$\int_0^h f(x) dx = \frac{h}{12} [6f(0) + 6f(h) + f'(0) - f'(h)] + \frac{h^5 f^{(4)}(\theta)}{720}$$

The composite formula is, therefore,

$$\begin{aligned} \int_0^{nh} f(x) dx &= \frac{h}{2} \{f(0) + 2f(h) + \cdots + 2f[(n-1)h] + f(nh)\} \\ &\quad + \frac{h}{12} [f'(0) - f'(nh)] + nh \frac{h^4 f^{(4)}(\theta)}{720} \end{aligned}$$

Thus only the end derivatives are used (this formula can also be found from the Euler-Maclaurin formula in Sec. 12.8).

PROBLEMS

Find the corresponding composite formula from the following problems:

20.5.1 18.10.4

20.5.2 17.4.1

20.5.3 17.5.3

20.5.4 17.5.4.

20.6 COMPOSITE OR HIGH-ACCURACY FORMULA?

How shall we choose between the two alternate choices of integration, a low-order composite formula or a single high-accuracy formula? The answer is both very simple and very complex. In principle it is a question of the rate of growth of the higher derivatives of the integrand [ignoring the kernel $K(x)$]. Section 14.7 discussed the growth of the higher derivatives for functions that have singularities in the finite part of the complex plane [recall that $1/(1+x^2)$ has singularities at $x = \pm i$] and showed that some of the high-order derivatives must behave like $n!$ or worse; otherwise the series would converge everywhere.

Since, except for a few elementary cases, the high-order derivatives are not available in closed form, it is usual to estimate them from a difference table. Of course this has its dangers, especially near a singularity. While we try to incorporate all the singularity we can into the kernel, there is usually some singularity left, not necessarily in the interval of integration, to cause trouble.

There is no magic number, but experience seems to indicate that for moderate engineering and scientific accuracies of, say, five decimal places, fourth- and fifth-order derivatives are about as far as it is wise to go in the absence of knowledge that the higher derivatives are not excessively large. On the other hand, experience with Gaussian integration indicates that good answers can be obtained from high-accuracy formulas. Apparently it is the equal spacing that is not favorable to the use of high-order derivatives in the Newton-Cotes, and similar, formulas.

20.7 GREGORY-TYPE FORMULAS

The trouble in estimating the value of the derivative occurring in the error term leads to sequential methods of estimating the accuracy of a formula. The use of several different spacings and the comparison of the resultant numbers provide a popular way of approaching the problem. One of the more popular methods is to start with a low-order formula with a known error-term structure, usually expressed in the Taylor-series form with many terms being kept, and then using

tighter and tighter spacings, build up a table of values of estimates of the integral. The next step is to eliminate pairwise the lowest-order derivative that appears in the error term; thus from n estimates we get $n - 1$ estimates with one of the derivatives eliminated. Then from these we eliminate the next derivative, etc. This is not the same as directly finding the most accurate formula you can, but it tends to avoid oscillation in the weights (the effective weights, once all the elimination is done) that occurs in the Newton-Cotes formulas.

It would seem, however, that the direct use of the Gregory, or Hamming-Pinkham, formula (which was invented for this purpose) would be better, and a number of experiments back up the theory. The approach through the differences allows almost independent estimates of *both* truncation and roundoff error, so that a comparison of the effects of using either a higher-order formula (using more differences) or more points can be made accurately.

The Gregory and Hamming-Pinkham formulas have an additional feature worth noting. When the integrand becomes flat at the ends of the interval, for example,

$$\int_{-\infty}^{\infty} e^{-x^2} dx$$

then the differences all approach zero and can be ignored.

Hartree¹ gives Table 20.7 for results obtained when computing

$$I = \int_0^{\infty} e^{-x^2} dx = h \left(1/2 + \sum_{k=1}^{\infty} e^{-h^2 k^2} \right)$$

The remarkable accuracy for $h = 1/2$ is unfortunately accompanied by a lack of knowledge of when it occurs or when h is too large; the error analysis of Gregory's formula is a difficult topic.

Table 20.7 COMPUTATION OF $\int_0^{\infty} e^{-x^2} dx^*$

I				
0.5	0.88622	69254	5	Correct to 11 decimals
0.6		69254	8	
0.7		69285		
0.8	0.88622	72808		
0.9	23	598		
1.0	32	0		
1.1	0.88674			

¹ See Hartree [19]. See also R. A. Fisher, *Phil. Trans. Roy. Soc. London, Ser. A*, 1922, for general h and error theory.

PROBLEMS

- 20.7.1 Make a flow diagram for the Gregory formula which involves the machine choice of when to stop using more differences. Discuss your reasoning for terminating as you do.

20.8 COMPOSITE INTERPOLATION

For interpolation the problem of whether to use one high-order polynomial for the whole interval or to use a sequence of lower-order polynomials for various intervals is exactly the same problem as when to use a single high-order integration formula or a different composite formula. The chief difference is that integration is a "smoothing" operation and is less vulnerable to the inevitable wiggles of a polynomial, and so there is more of a tendency to go to composite formulas when interpolating. However, among other difficulties in the interpolation process, when we shift from one interval to the next, we have the situation that there are sharp corners.

Spline interpolation was designed to cope with this problem, and for most practical purposes spline interpolation means cubic spline interpolation. In cubic spline interpolation the function and the first and second derivatives are required to agree at the ends of the intervals where one cubic meets the next. Thus we have a continuous second derivative throughout the whole range of interpolation.

Spline interpolation is useful in many situations which require a smooth interpolating function through given data. Examples include interpolating data to drive automatic milling machines, many draughting applications where "smoothness" is highly desirable, and in computer science, graphical output of smooth curves from given data.

Spline interpolation is a large, well developed field, and we can, per usual, give only a brief introduction [2].

20.9 THE CUBIC SPLINE EQUATIONS¹

Let the ordinates y_i be given at x_i ($i = 1, \dots, n$), respectively. Let $h_i = x_{i+1} - x_i$ denote the mesh-point spacing. Let $y(x)$ be an interpolation curve through these points and define y'_i and y''_i as the first and second derivatives, respectively, of $y(x)$ at $x = x_i$. Let $y(x)$ be expressed in piecewise fashion as

$$y(x) = f_i(x) \begin{cases} \text{for } x_i \leq x < x_{i+1} & i = 1, \dots, n-2 \\ \text{for } x_{n-1} \leq x \leq x_n & i = n-1 \end{cases} \quad (20.9.1)$$

¹ Much of this material is taken from a Bell Telephone Laboratories memorandum by D. G. Schweikert.